



UNIVERSITÀ DI PARMA

Camera Models

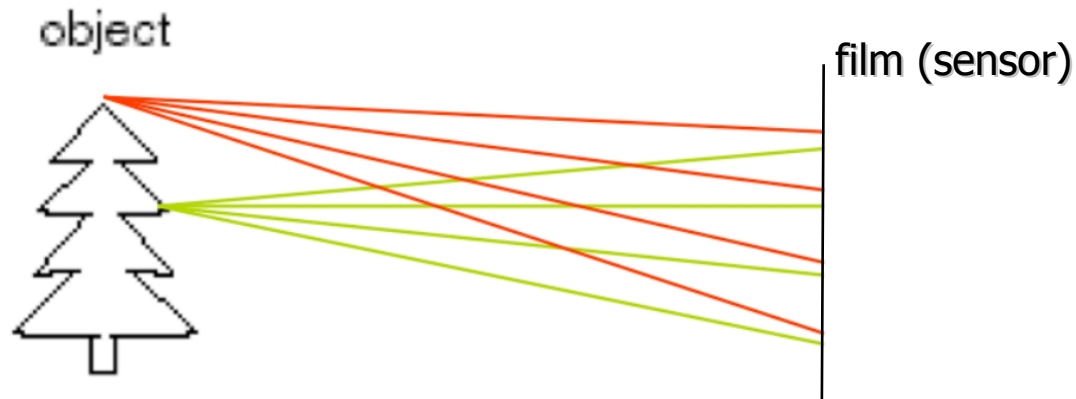
- Pin Hole Camera
- Lenses
- Pin-Hole Camera Geometry

Courtesy of CS231A · *Computer Vision: from 3D reconstruction to recognition*, Prof. Silvio Savarese – Stanford University

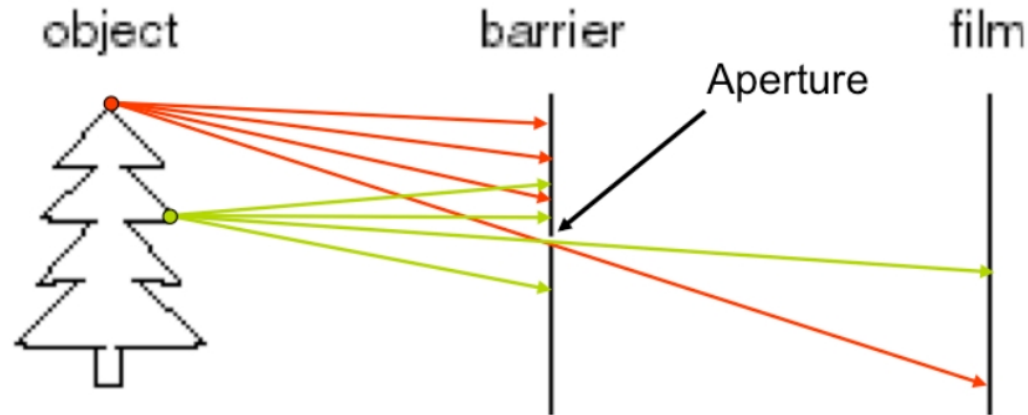
- Since now we discussed image processing
- Namely, we saw fundamental techniques to process a 2D matrix...
- How that image is created?
- What is the relation, if any, to the 3D world?
- During this lesson we will try to answer to those questions



- Too much simple camera
 - A sensitive film in front of an object
 - What is the result?

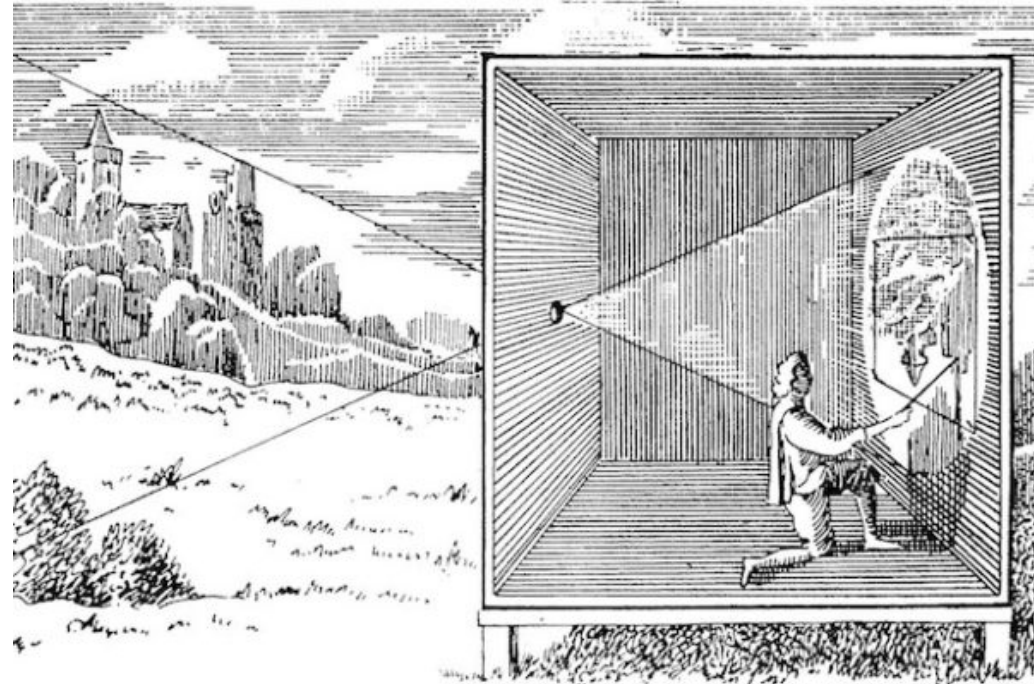


- We can add a barrier with a very small hole
 - So-called pin-hole or aperture
 - Now, only one ray hits the film in a given position

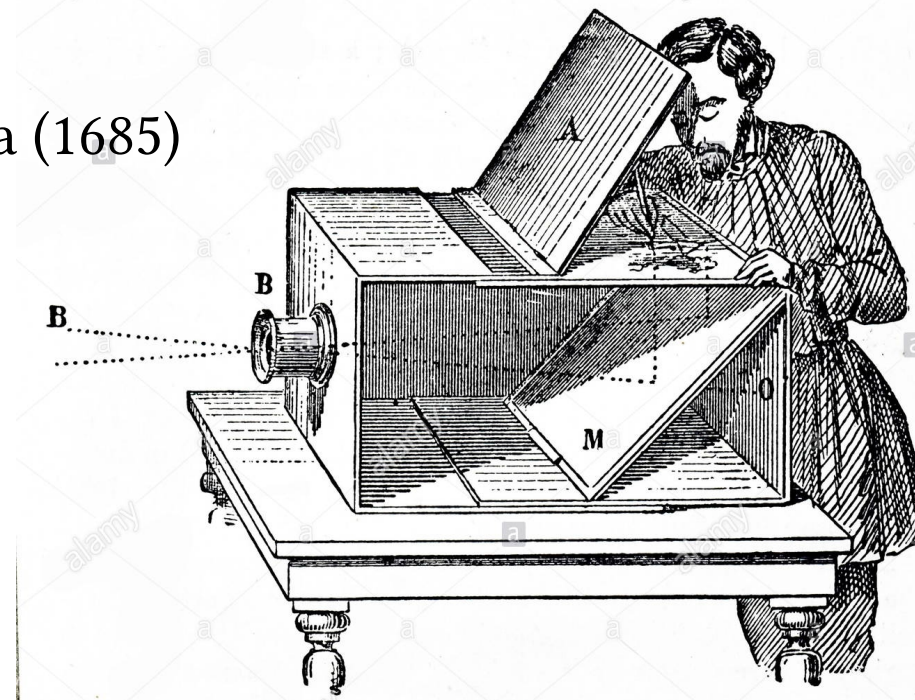




- Milestones
 - Leonardo da Vinci's *Camera Obscura* (1502)



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 - Johan Zahn: first portable camera (1685)

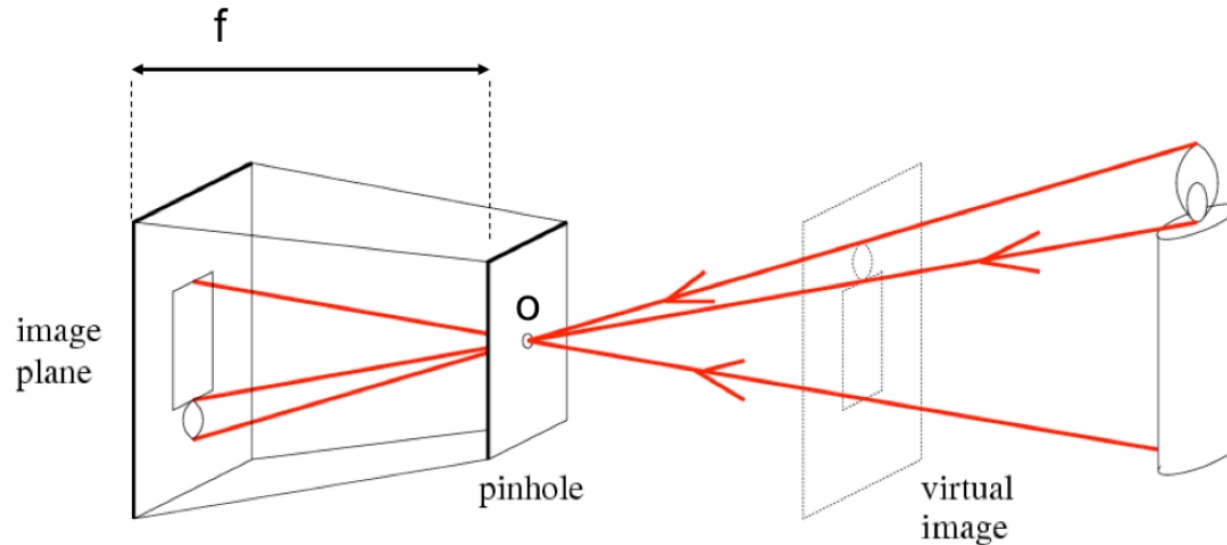


- Milestones
 - Leonardo da Vinci's *Camera Obscura* (1502)
 - Johan Zahn: first portable camera (1685)
 - Joseph Nicéphore Niépce: first photo (1822)

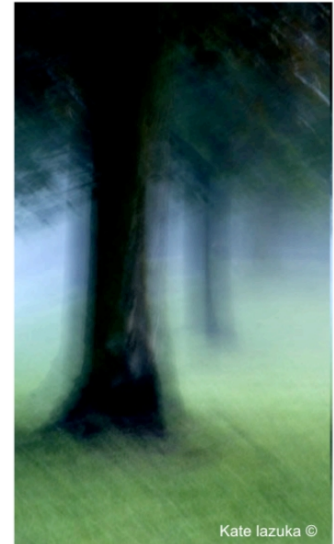
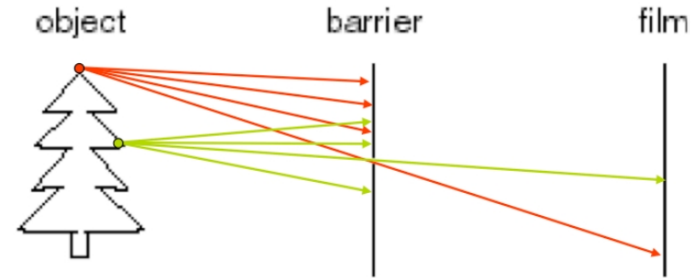


Pin-Hole Camera

- $f \rightarrow$ focal length
- $o \rightarrow$ pin-hole, aperture (center of the lens)

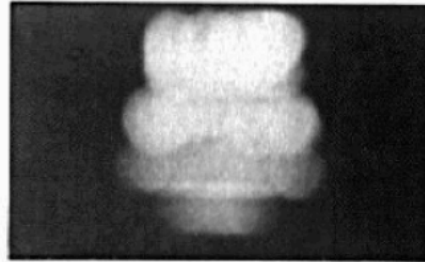
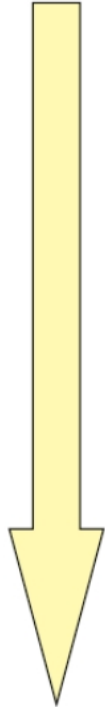


- The larger the pin-hole, the greater the number of rays
 - More rays $\rightarrow$ More light energy
 - More rays $\rightarrow$ Blurred image



Camera Aperture size

Aperture Size
decrease



2 mm



1 mm

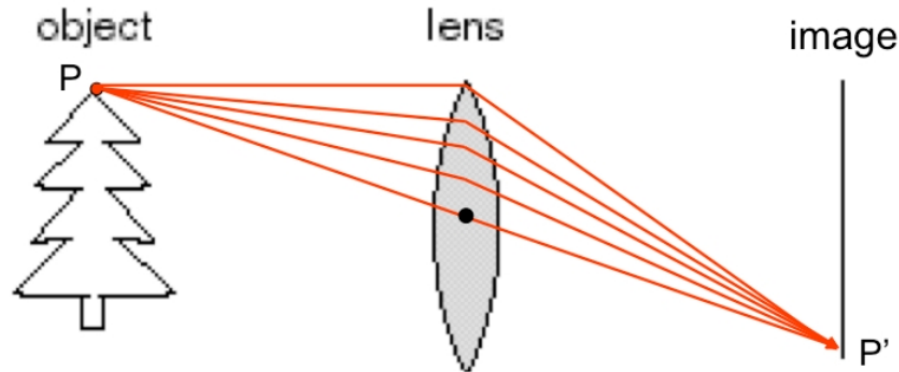


0.6mm

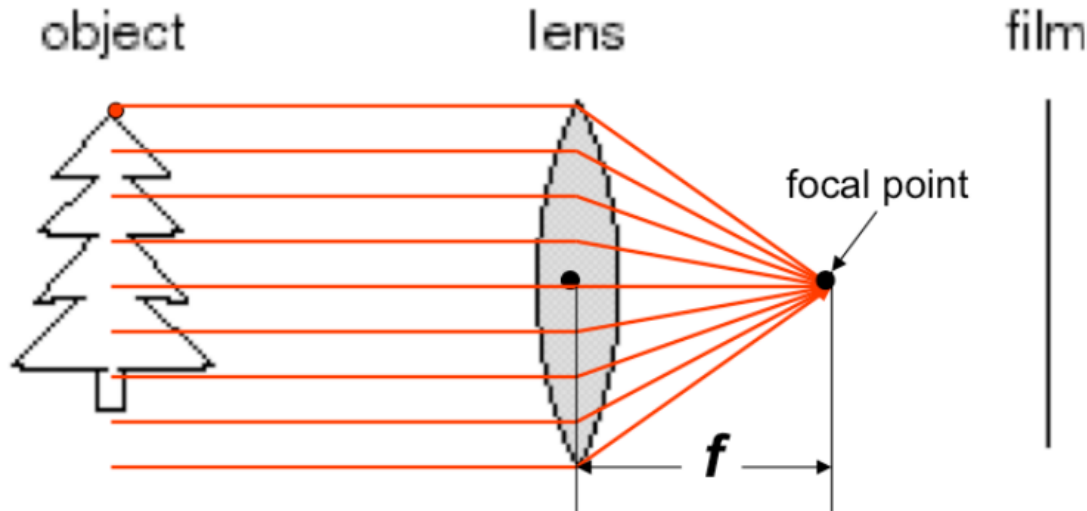


0.35 mm

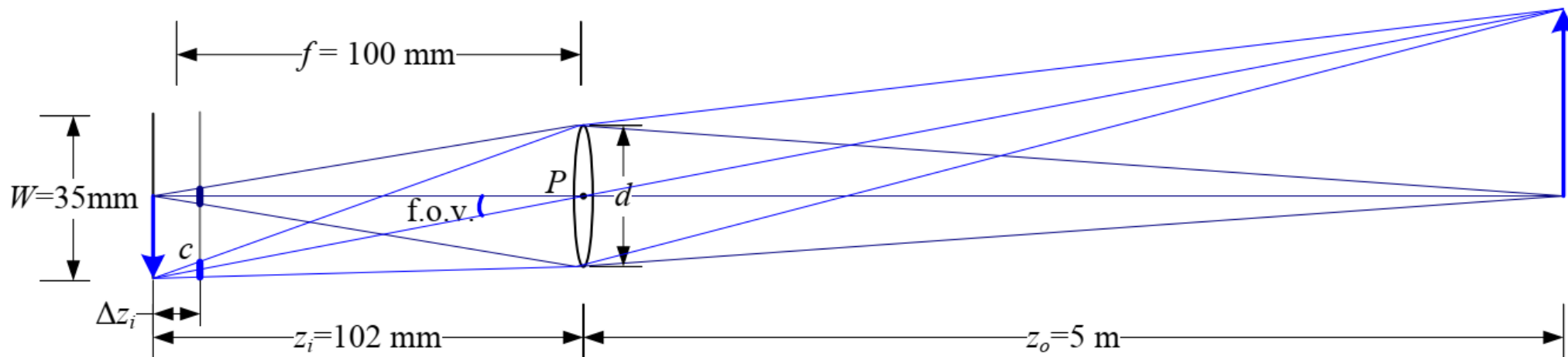
- Light vs Precision
 - Actual pin-holes are barely used
- Lenses are much more comfortable (some issues anyway)
 - We can intercept more rays coming from same 3D point



- Parallel light rays converge in a specific point
 - Focal point at distance f from the center of the lens
- Only the ray passing through the center of the lens is not deviated

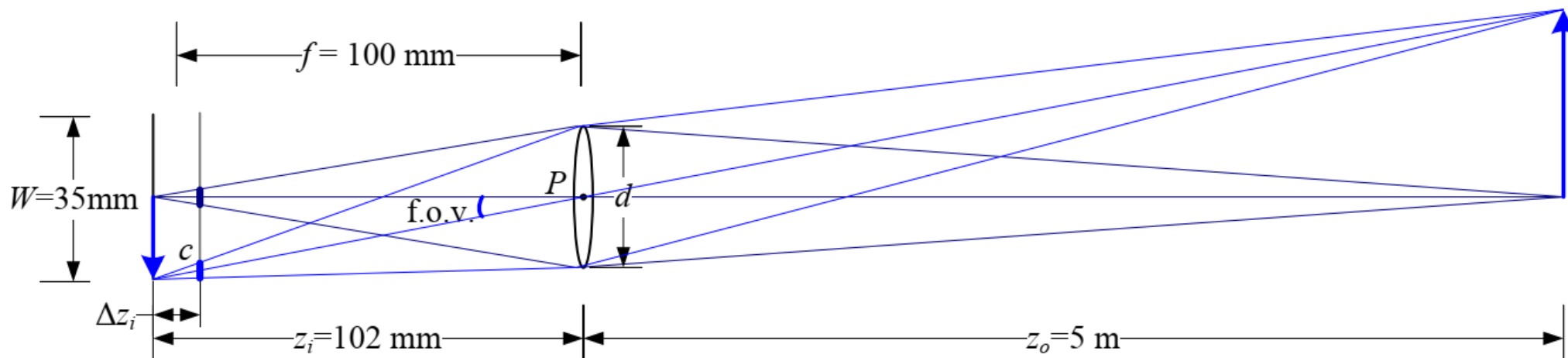


- Only rays emitted from a given plane points in the world converge to points that lie on a specific plane
- The following formula can be used:
$$\frac{1}{f} = \frac{1}{z_i} + \frac{1}{z_o}$$
 - Varying z_o means that also z_i is modified

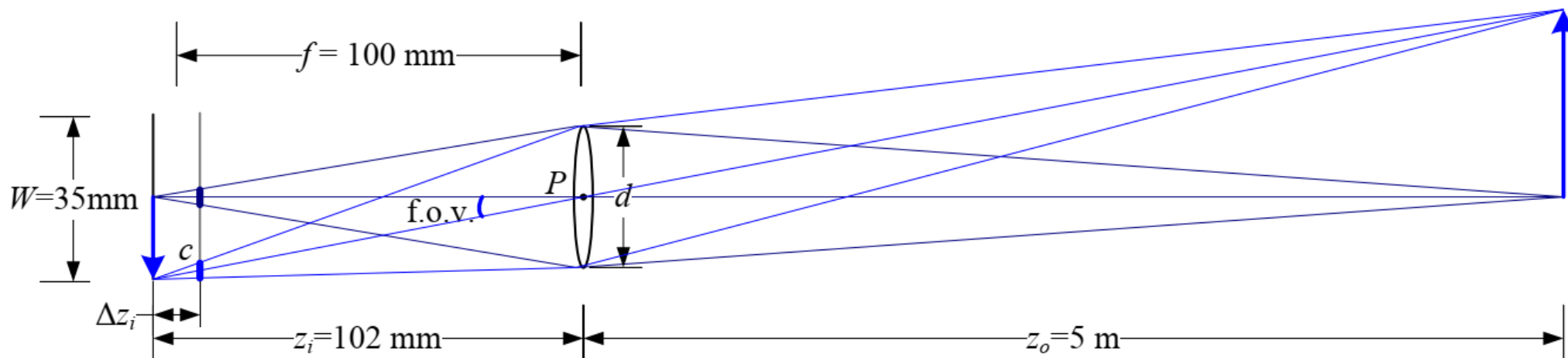


- Therefore the distance between lens and sensor give us the *perfect* focus distance
- For other points we have a **circle of confusion**

$$\frac{1}{f} = \frac{1}{z_i} + \frac{1}{z_o}$$

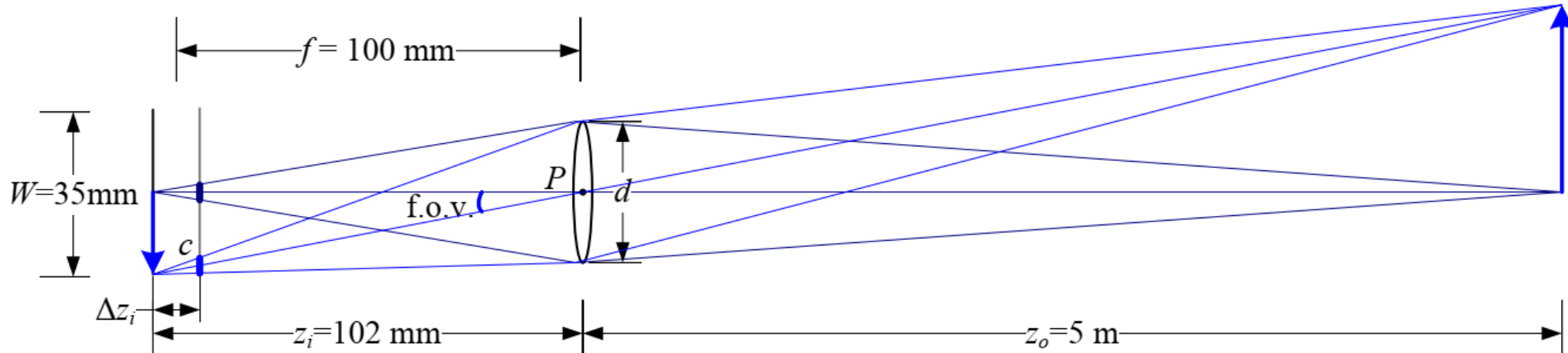


- Anyway in the real world sensor elements have a finite size
- Then we can consider a small portion of the world to be sufficiently in focus: **Shallow Depth of Field**
- Given the $1/z_0$ for z_0 sufficiently large this can mean up to ∞



- The depth of field depends on the f-number or focal ratio ($f/\#$)
- The higher the f-number, the larger the depth of field

$$f/\# = \frac{f}{d}$$

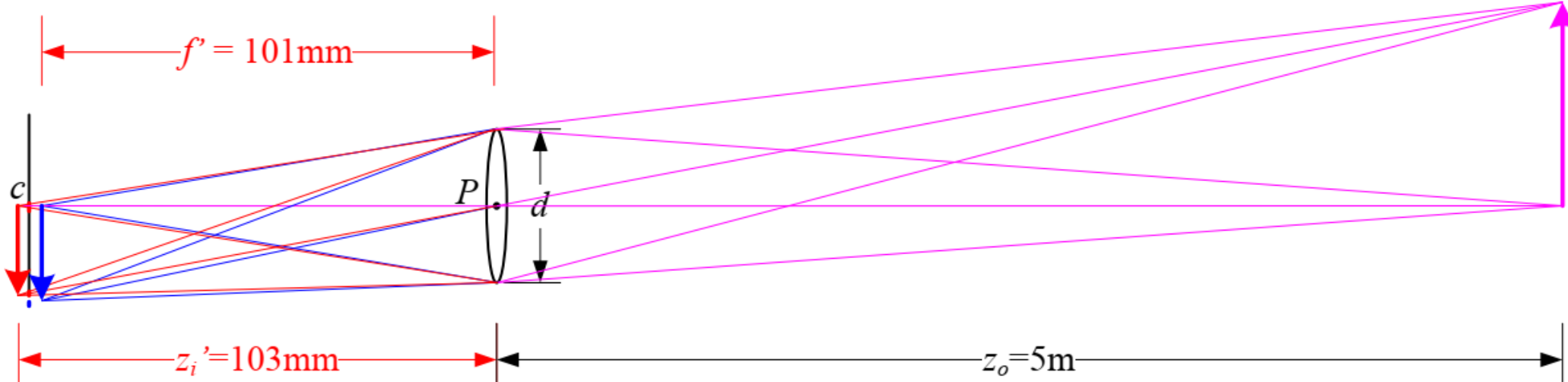


Shallow Depth of Field

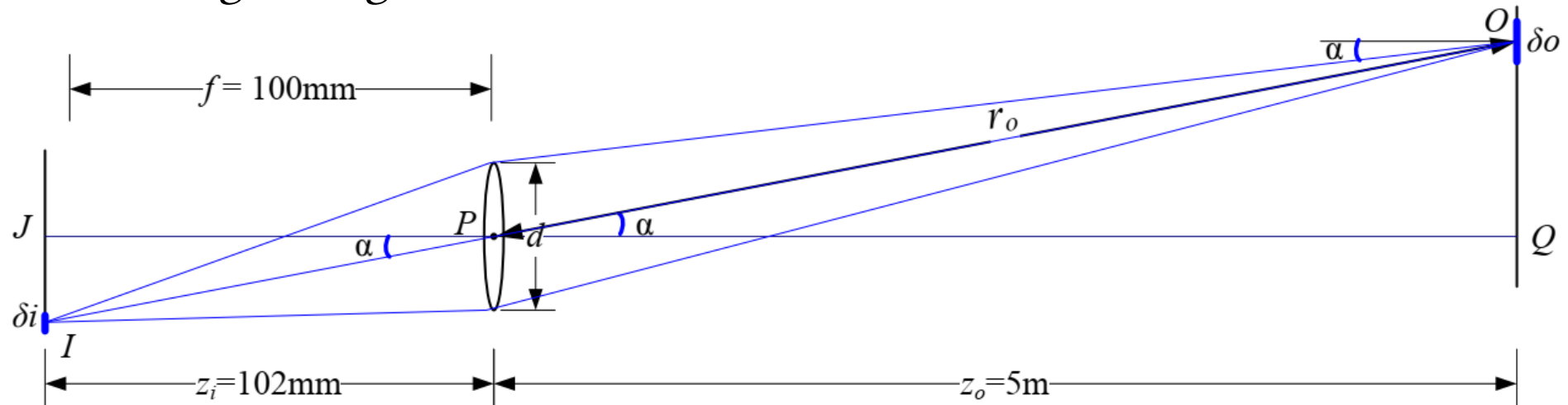
- Example of shallow depth of field



- Refractive index depends on wavelength
- Different colors are then projected in different positions
- **Chromatic Aberration**



- Areas in the borders/corners are typically darker...
- Distance from optical axis affects energy
 - Vignetting



Distortion

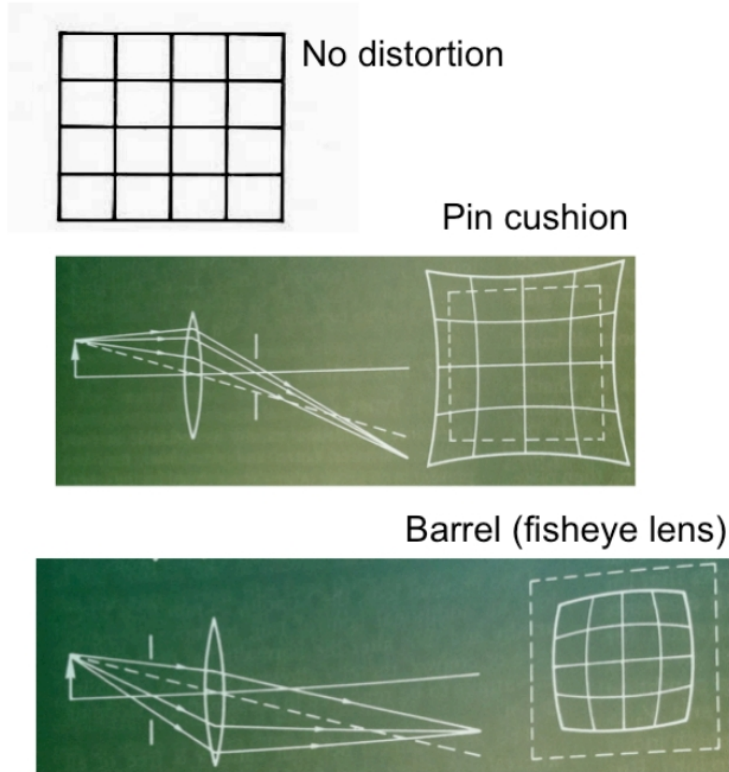
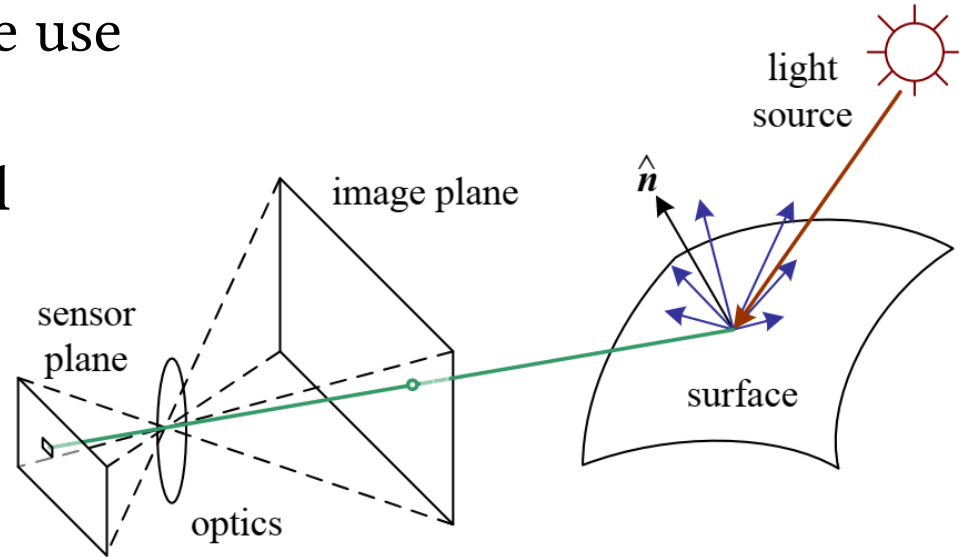


Image magnification decreases
with distance from the optical axis

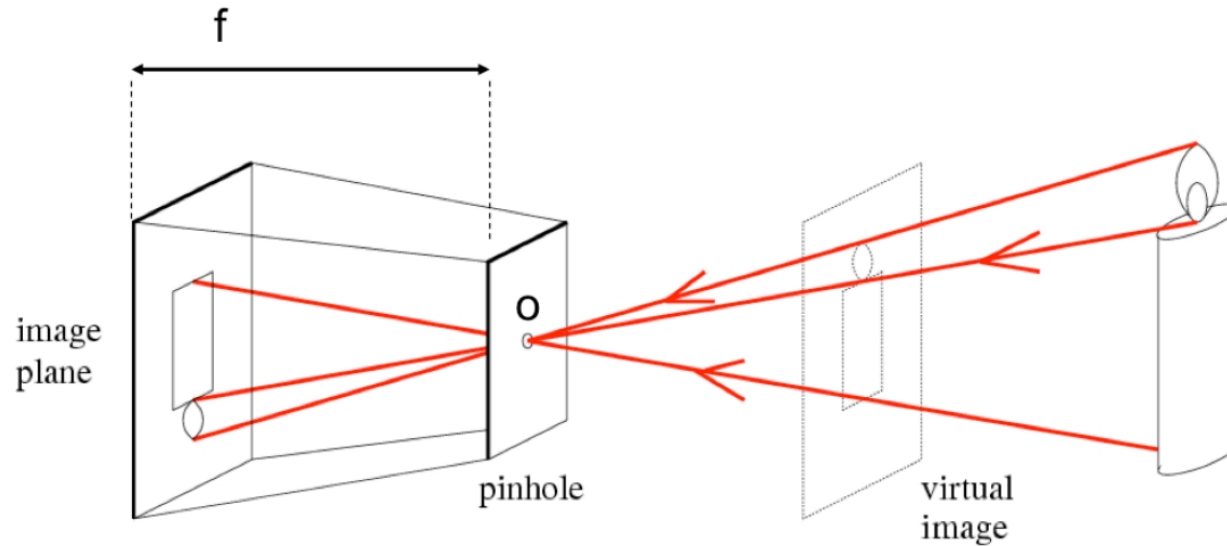
- Distortion effect is much more evident in lateral areas

- We saw many issues related to the use of a lens
- This also affects the camera model
- Anyway in the following we will simply assume to have:
 - Thin lenses
 - Small angles of view
 - No or compensated chromatic aberration
 - No or rectified distortion effects

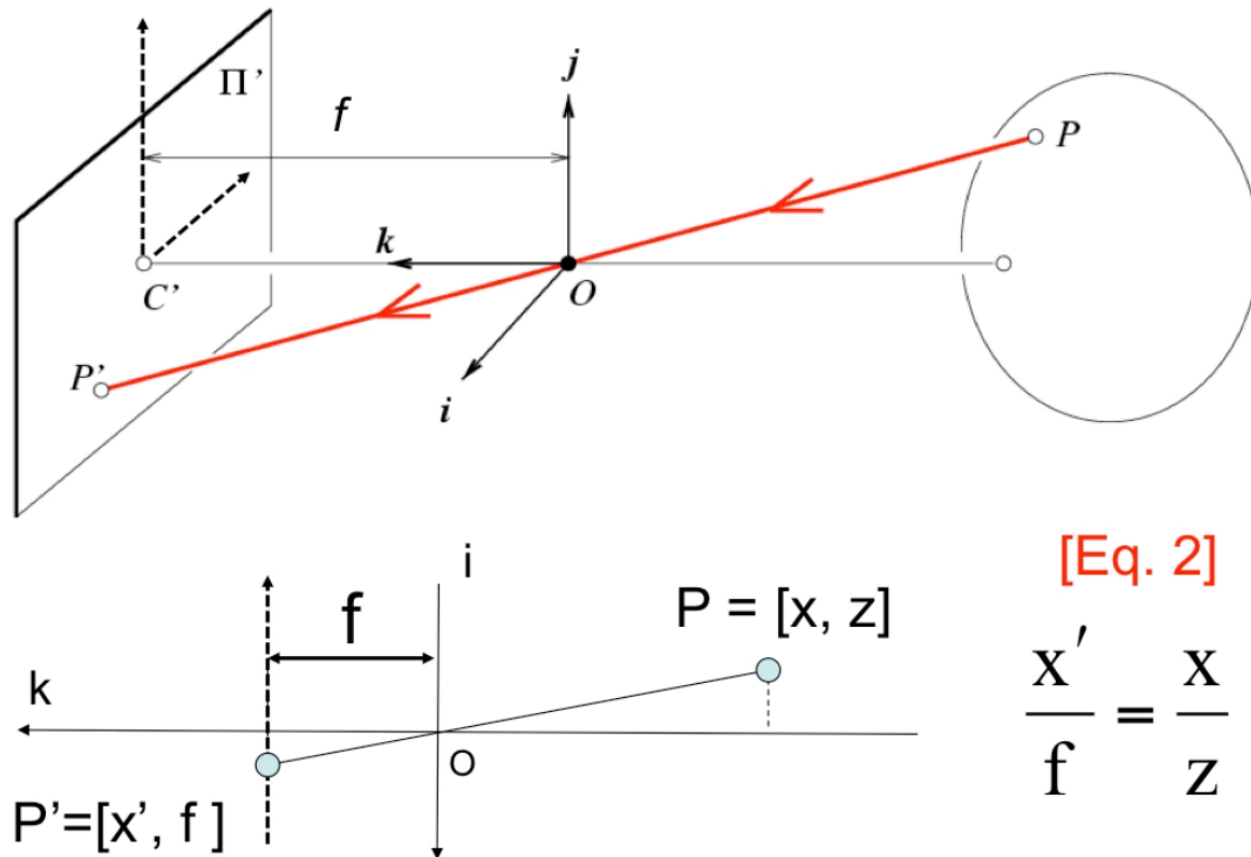


Pin-Hole Camera

- $f \rightarrow$ focal length
- $o \rightarrow$ pin-hole, aperture (center of the lens)



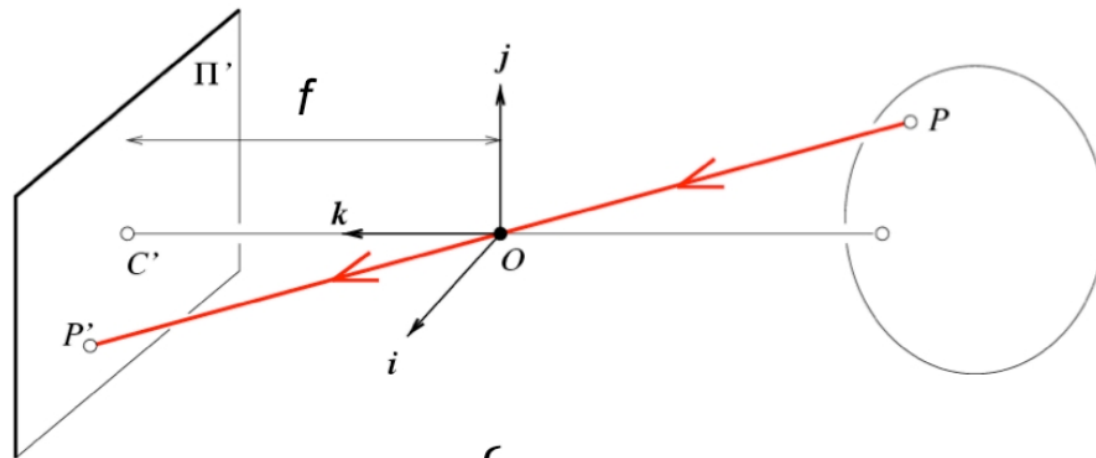
Pin-Hole Camera



[Eq. 2]

$$\frac{x'}{f} = \frac{x}{z}$$

Pin-Hole Camera: perspective transformation



$$P = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \rightarrow P' = \begin{bmatrix} x' \\ y' \end{bmatrix}$$

[Eq. 1]

$$\begin{cases} x' = f \frac{x}{z} \\ y' = f \frac{y}{z} \end{cases} \quad \mathbb{R}^3 \xrightarrow{E} \mathbb{R}^2$$

f = focal length
 o = center of the camera

Image & Sensor Planes

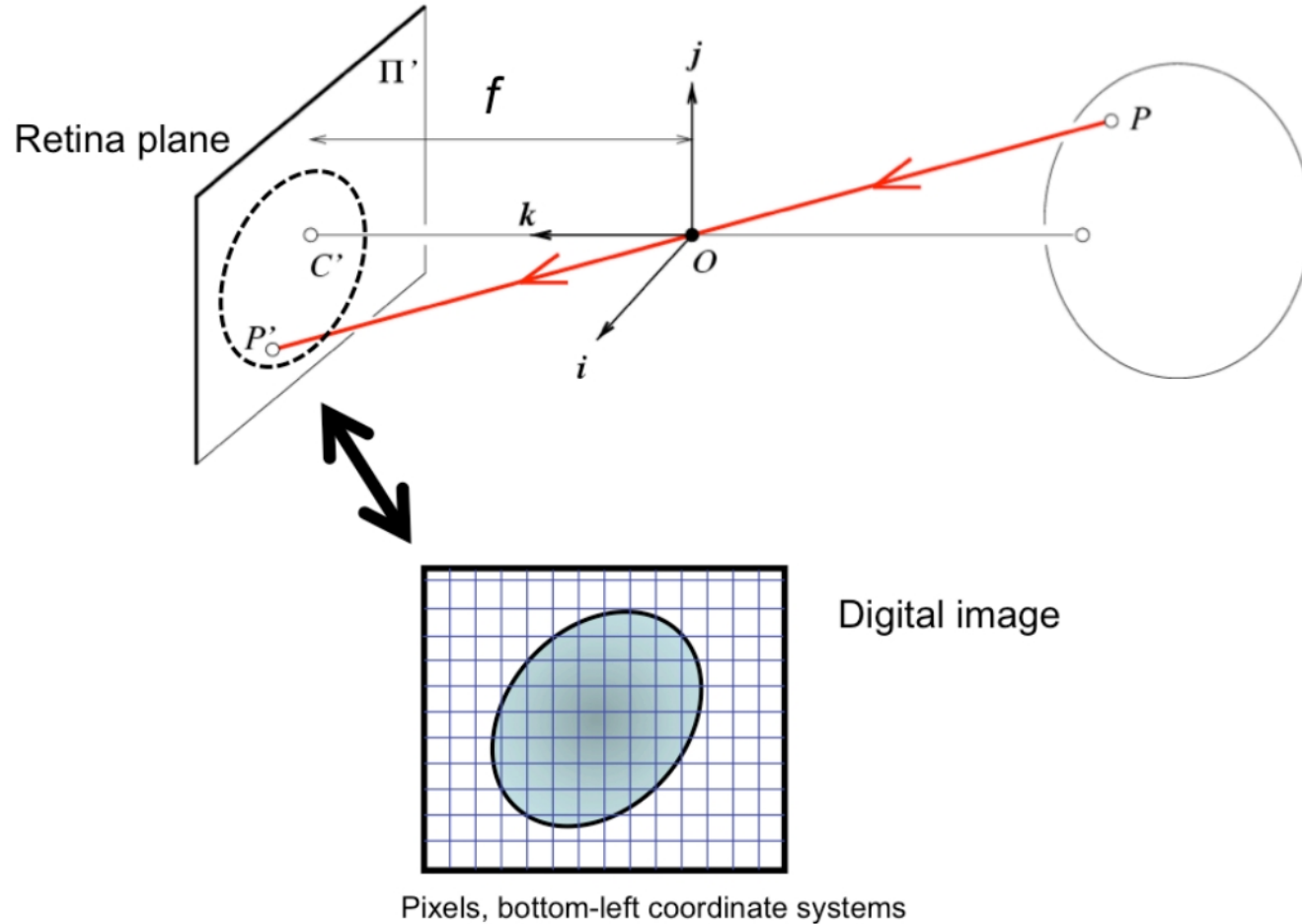
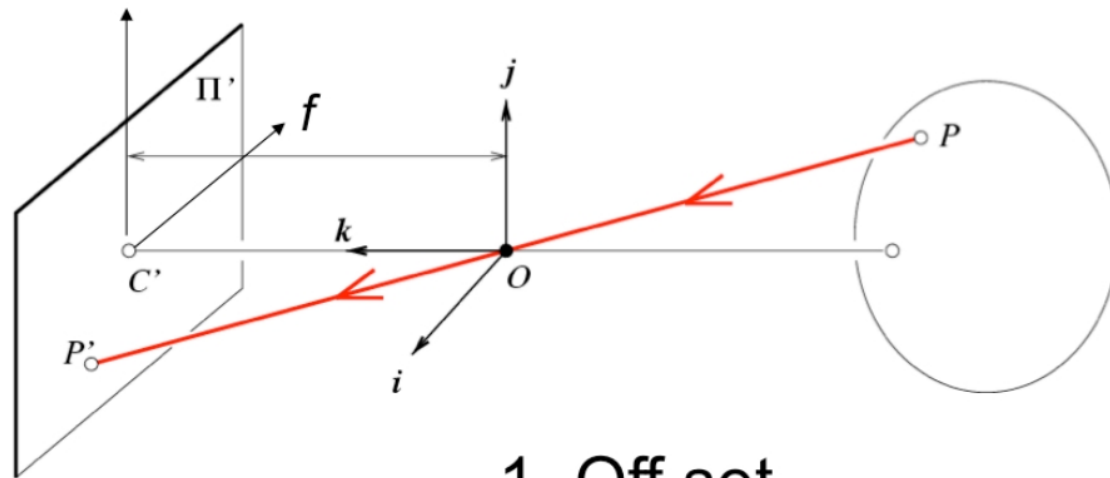
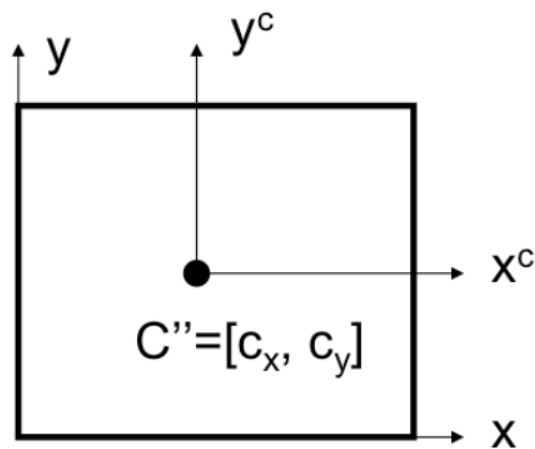


Image & Sensor Planes → Origin



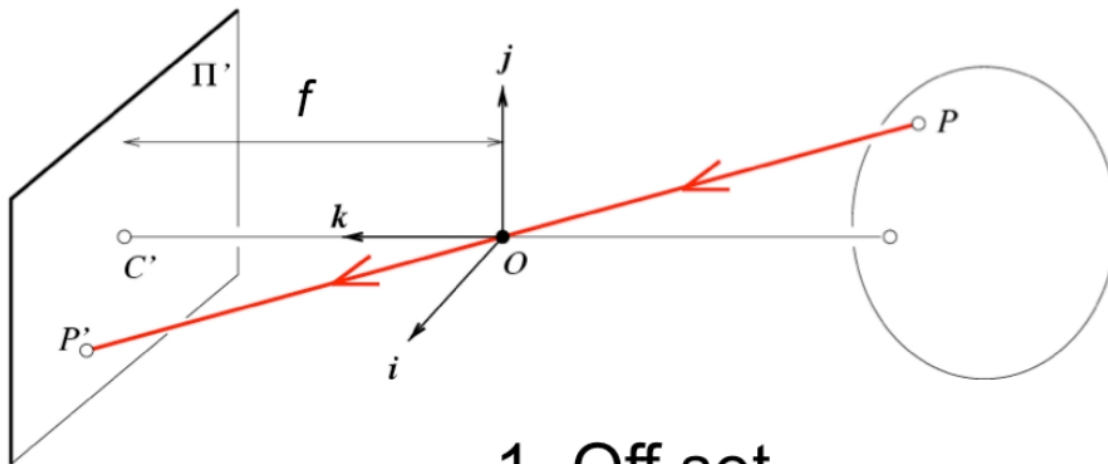
1. Off set



$$(x, y, z) \rightarrow \left(f \frac{x}{z} + c_x, f \frac{y}{z} + c_y\right)$$

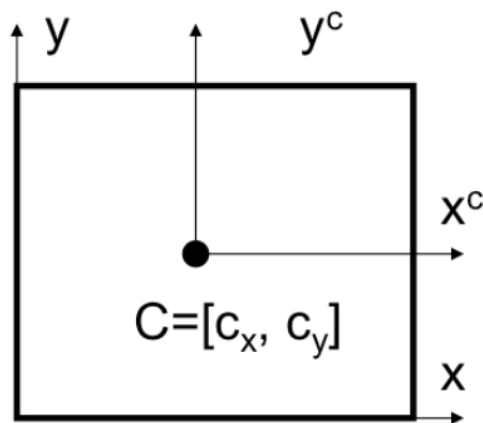
[Eq. 5]

Image & Sensor Planes → Pixel size



1. Off set

2. From metric to pixels



$$(x, y, z) \rightarrow \underbrace{\left(f \frac{k}{z}\right)}_{\alpha} + c_x, \underbrace{\left(f \frac{l}{z}\right)}_{\beta} + c_y \quad [\text{Eq. 6}]$$

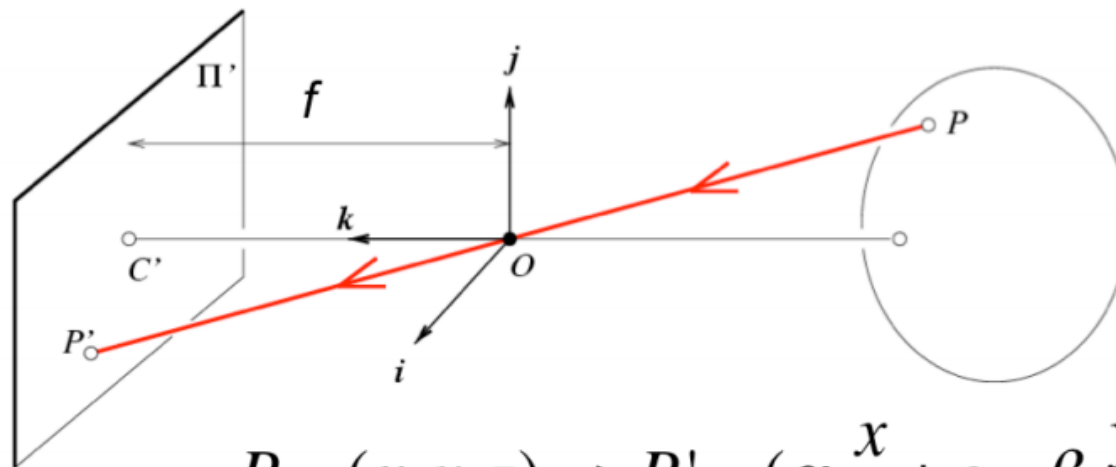
Units: k, l : pixel/m

Non-square pixels

f : m

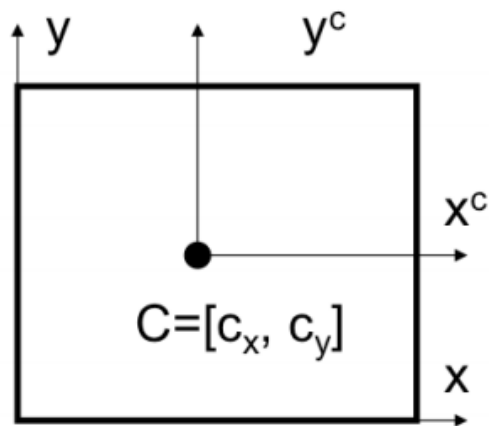
α, β : pixel

Non Linear Transformation



$$P = (x, y, z) \rightarrow P' = \left(\alpha \frac{x}{z} + c_x, \beta \frac{y}{z} + c_y \right)$$

[Eq. 7]



- Is this a linear transformation?
No — division by z is nonlinear
- Can we express it in a matrix form?

- The non linearity can be solved using Homogeneous Coordinates
- HC are an augmented representation of points
- We add another “coordinate”, i.e. $\mathbb{R}^n \rightarrow \mathbb{R}^{n+1}$
- In 2D space $P=(x, y)$ can be represented as $P=(x, y, 1)$
 - Or more generally as (kx, ky, k)
 - The third value can be considered as a scale factor

Homogeneous vs Euclidean

- Conversions are simple:
- Euclidean $\rightarrow$ Homogeneous

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

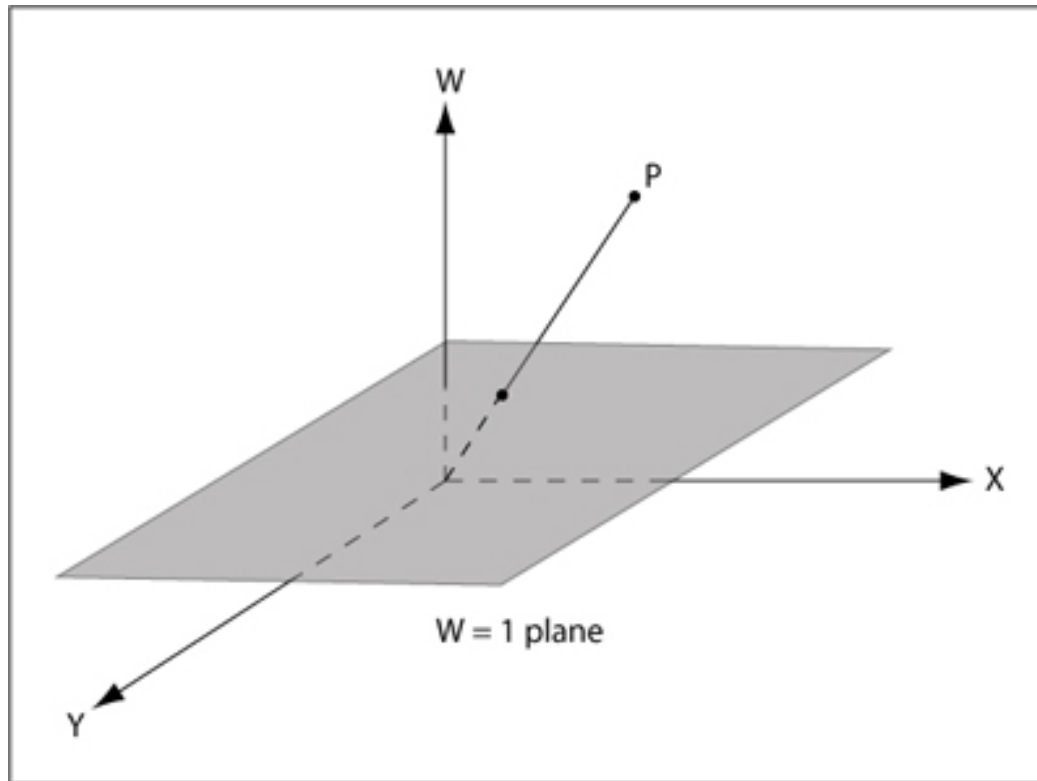
$$(x, y, z) \Rightarrow \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

- Homogeneous $\rightarrow$ Euclidean

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w)$$

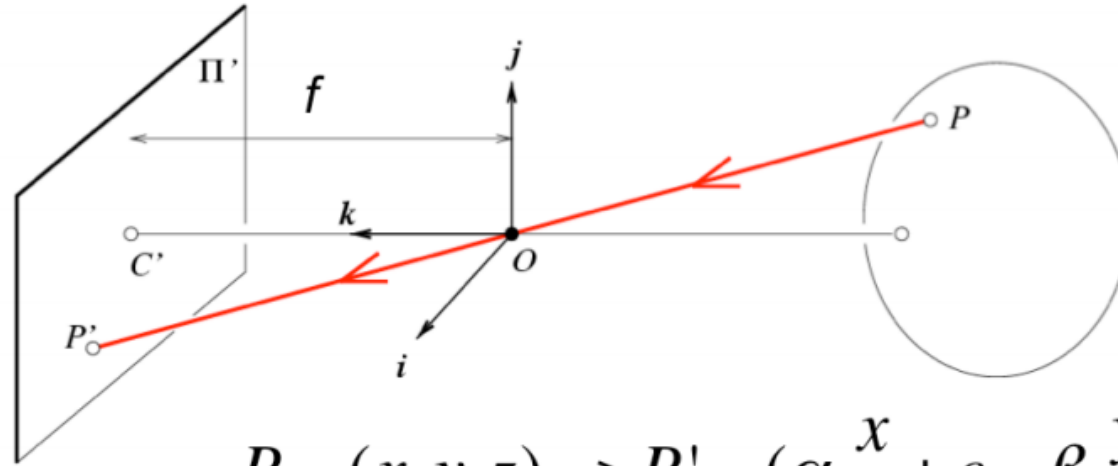
$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \Rightarrow (x/w, y/w, z/w)$$

- A geometric interpretation for HC can be given as follows



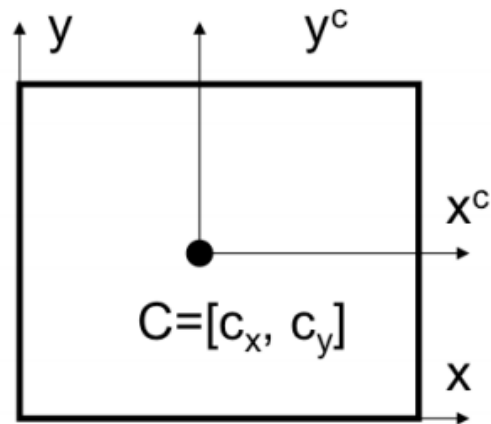
- Three main reasons to use homogeneous coordinates
 - Simple representation of points and lines (no special cases)
 - Homogenous space contains more points than Euclidean one!
 - $(x,y,0)$
 - Simple representation of Euclidean Transformations
 - Translation
 - Scale
 - Rotation
 - Simple representation of perspective projections

Non Linear Transformation (again)



$$P = (x, y, z) \rightarrow P' = \left(\alpha \frac{x}{z} + c_x, \beta \frac{y}{z} + c_y \right)$$

[Eq. 7]



- Is this a linear transformation?
No — division by z is nonlinear
- Can we express it in a matrix form?

- $P \rightarrow P'$ projection becomes $P_h \rightarrow P_h'$
- The $P=[x \ y \ z]$ in the 3D space is $P_h=[x \ y \ z \ 1]$ in Homogeneous reference system
- P' was computed as $[\alpha(x/z)+c_x \ \beta(y/z)+c_y]$
- P_h' can be then $[\alpha x+c_x z \ \beta y+c_y z \ z]$
- Do you see how to express $P_h \rightarrow P_h'$ as matrix product?

Perspective Linear Transformation

$$P_h' = \begin{bmatrix} \alpha x + c_x z \\ \beta y + c_y z \\ z \end{bmatrix} = \begin{bmatrix} \alpha & 0 & c_x & 0 \\ 0 & \beta & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \quad \text{--- } P_h \quad \text{[Eq.8]}$$

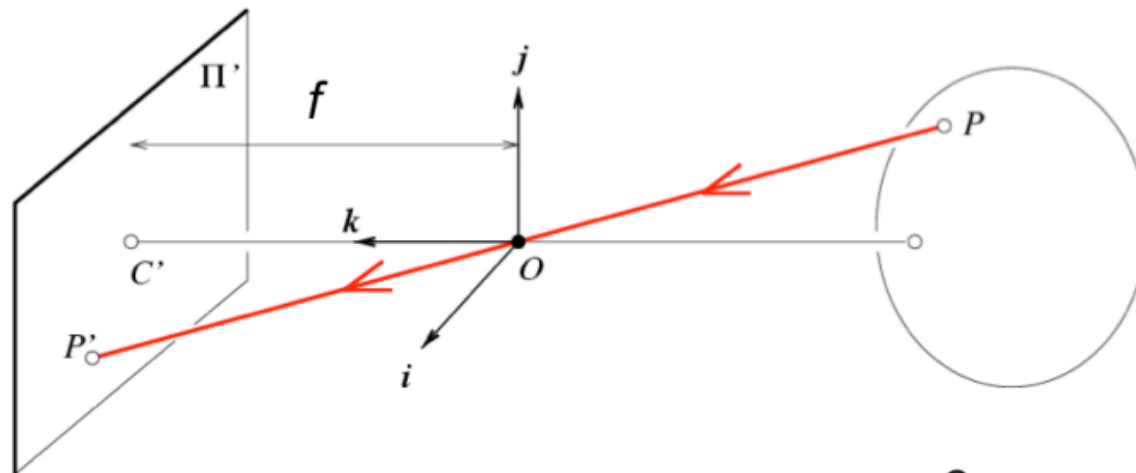
Homogenous

Euclidian

$$\underbrace{P_h'}_{\text{Homogenous}} \rightarrow \underbrace{P' = \left(\alpha \frac{x}{z} + c_x, \beta \frac{y}{z} + c_y \right)}_{\text{Euclidian}}$$

$$M = \begin{bmatrix} \alpha & 0 & c_x & 0 \\ 0 & \beta & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

The Intrinsic Matrix



Camera
matrix K

[Eq.9]

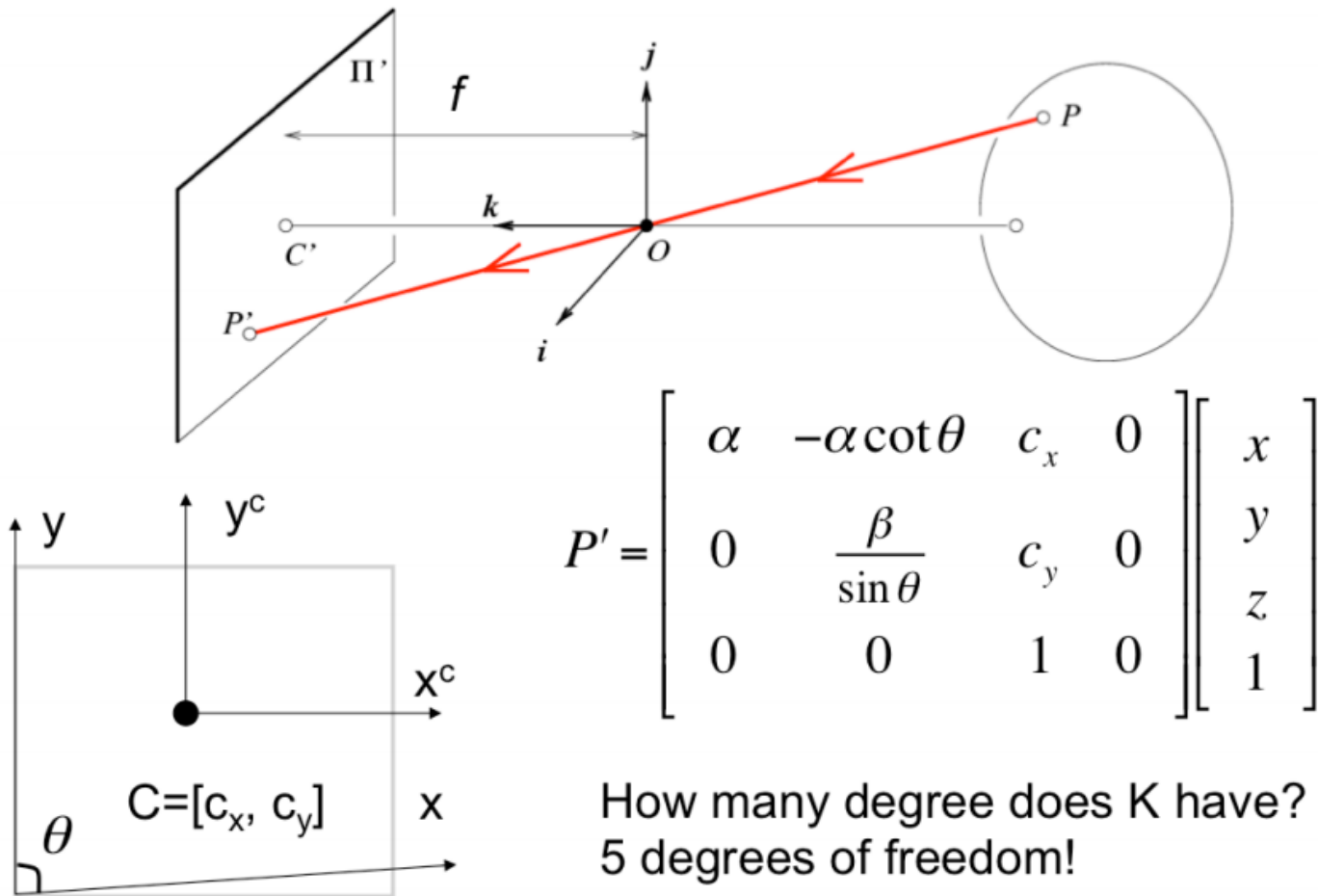
$$P' = M P$$

$$= K \begin{bmatrix} I & 0 \end{bmatrix} P$$

$$P' = \begin{bmatrix} \alpha & 0 & c_x & 0 \\ 0 & \beta & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

- Not enough, we have to consider skewness!
- Sometimes, the 2D image plane is not a rectangle but rather is skewed
 - i.e. the angle between the image axis is not 90 degrees.
- Another transformation needs to be carried out to go from the rectangular plane to the skewed plane
- No demonstration

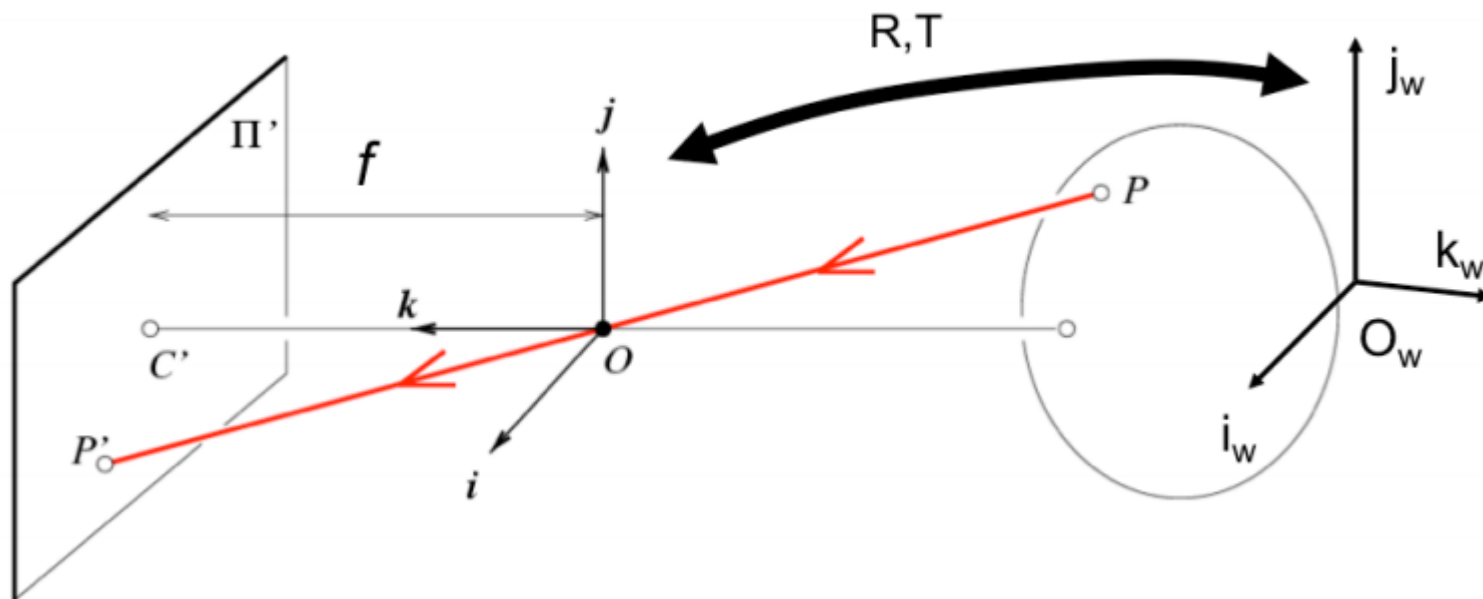
Skewness



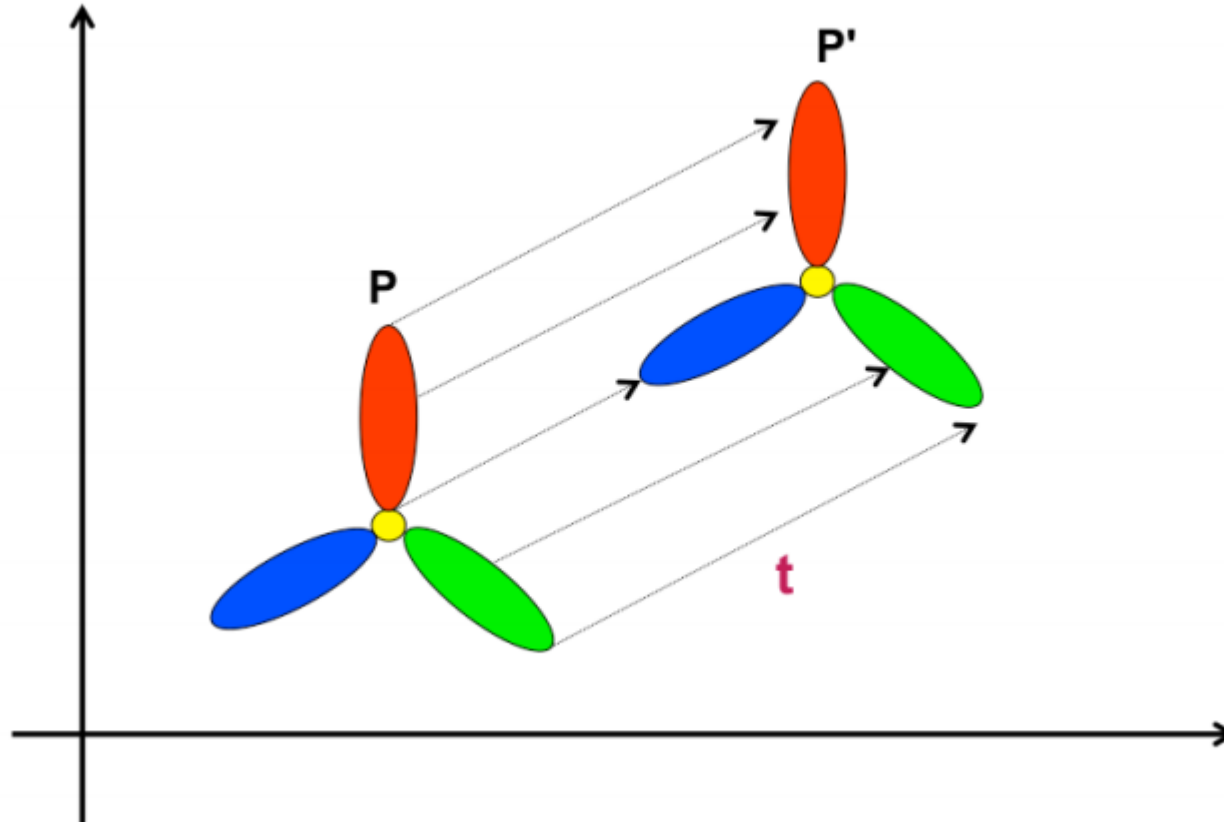
- So far we adopted a camera centric approach
 - Reference system centered on the pin-hole
 - All parameters are camera dependent only
 - No external world
- $K \rightarrow$ **Camera Intrinsic Matrix**

Introducing the external world...

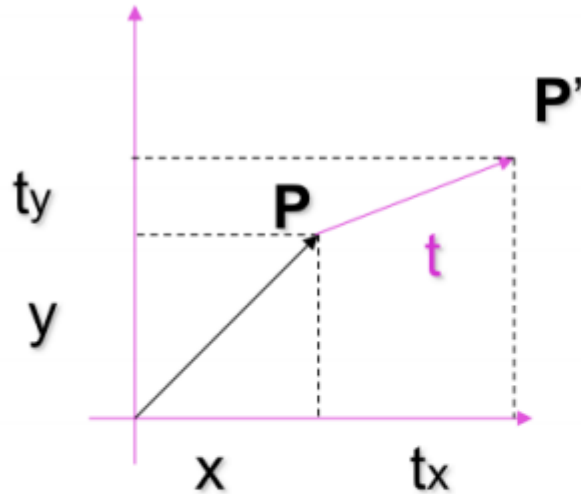
- Having a camera reference system is a bit limiting
- Usually a different reference system is used
- We need an additional transformation



Review: 2D translation



Review: 2D translation

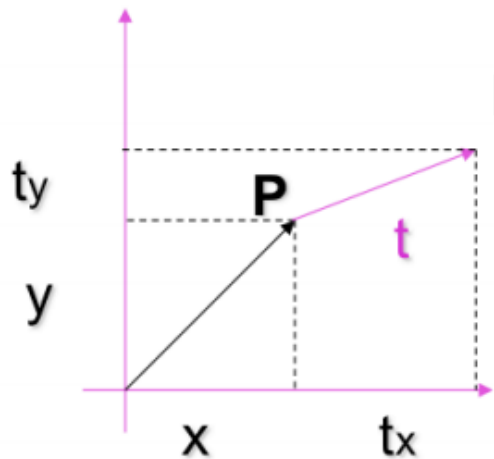


$$\mathbf{P} = (x, y)$$

$$\mathbf{t} = (t_x, t_y)$$

$$\mathbf{P}' = \mathbf{P} + \mathbf{t} = (x + t_x, y + t_y)$$

Review: homogeneous 2D translation

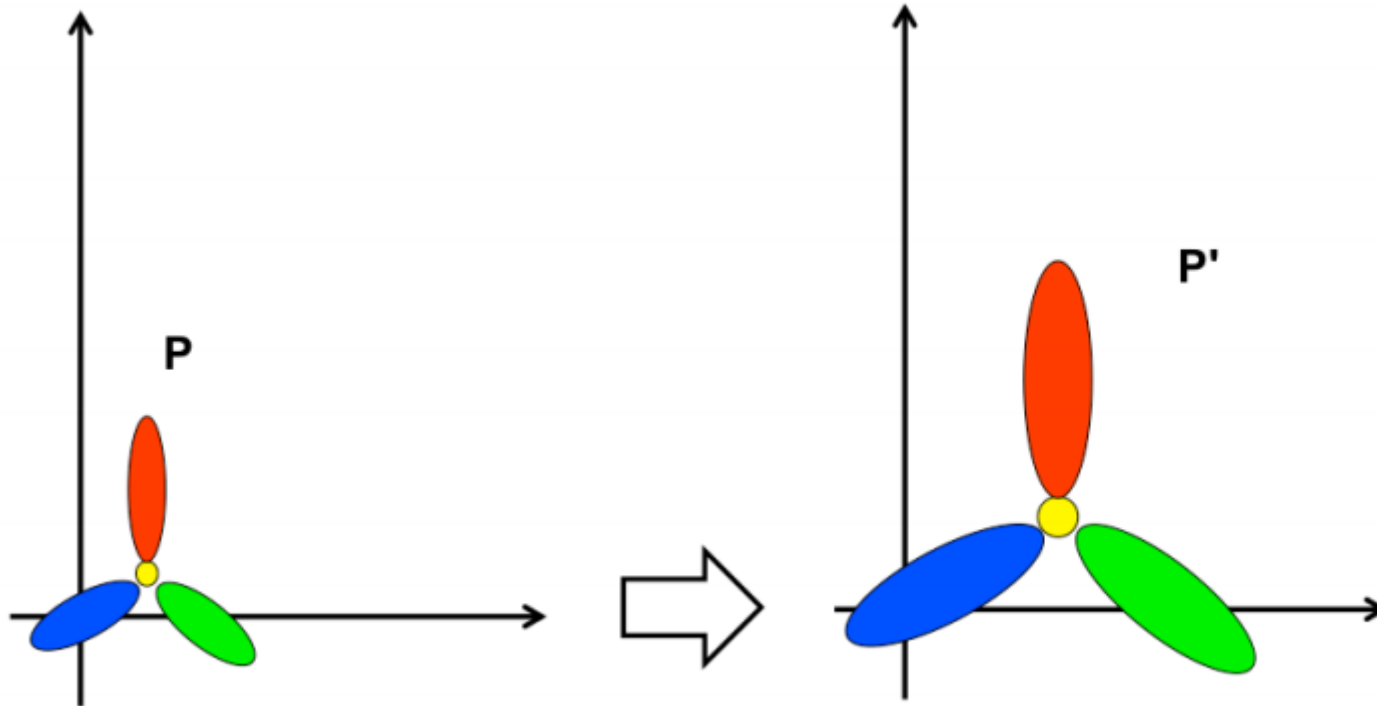


$$\mathbf{P} = (x, y) \rightarrow (x, y, 1)$$

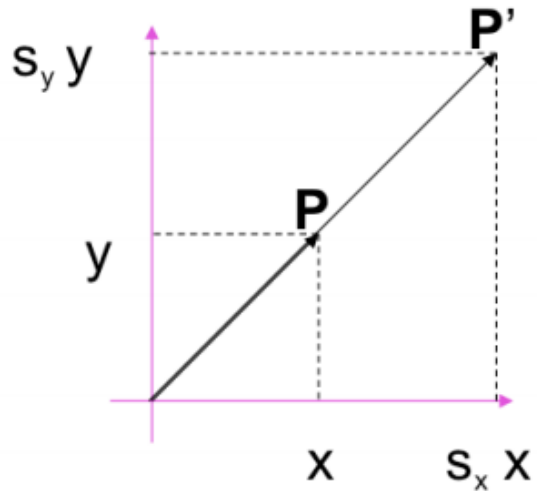
$$\mathbf{P}' \rightarrow \begin{bmatrix} x + t_x \\ y + t_y \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} \mathbf{I} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \mathbf{T} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Review: 2D scaling



Review: homogeneous 2D scaling

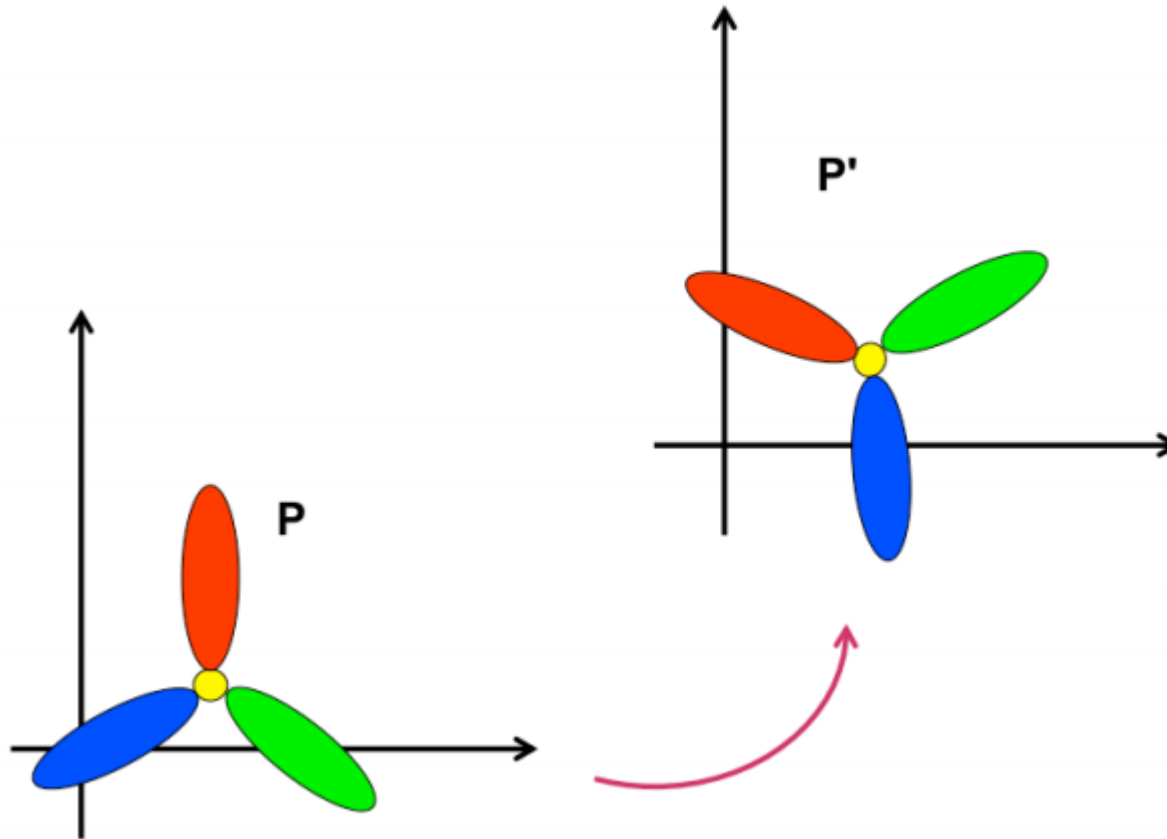


$$\mathbf{P} = (x, y) \rightarrow \mathbf{P}' = (s_x x, s_y y)$$

$$\mathbf{P} = (x, y) \rightarrow (x, y, 1)$$

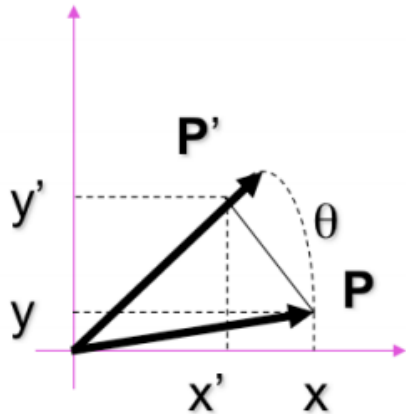
$$\mathbf{P}' \rightarrow \begin{bmatrix} s_x x \\ s_y y \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{S}} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{S}' & \mathbf{0} \\ \mathbf{0} & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \mathbf{S} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Review: 2D rotation



Review: 2D rotation

- Rotate around the z axis by Θ



$$x' = \cos \theta x - \sin \theta y$$

$$y' = \cos \theta y + \sin \theta x$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\mathbf{P}' = \mathbf{R} \mathbf{P}$$

How many degrees of freedom? 1

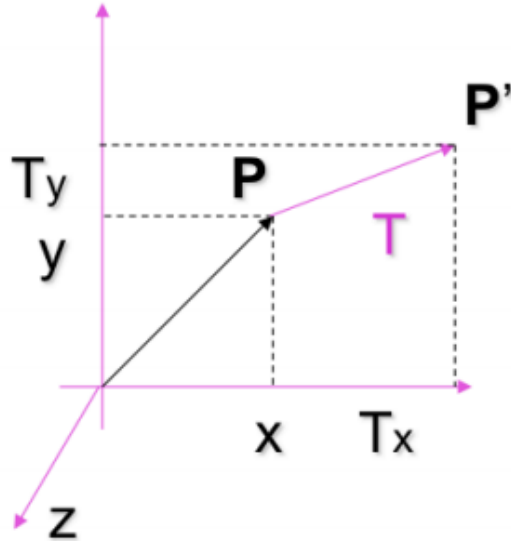
$$\mathbf{P}' \rightarrow \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Review: put everything together

$$\begin{aligned}\mathbf{P}' &\rightarrow \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta & -\sin \theta & t_x \\ \sin \theta & \cos \theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{S} & \mathbf{0} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \boxed{\begin{bmatrix} \mathbf{R} \mathbf{S} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix}} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}\end{aligned}$$

If $s_x = s_y$, this is a similarity transformation

Review: 3D translation

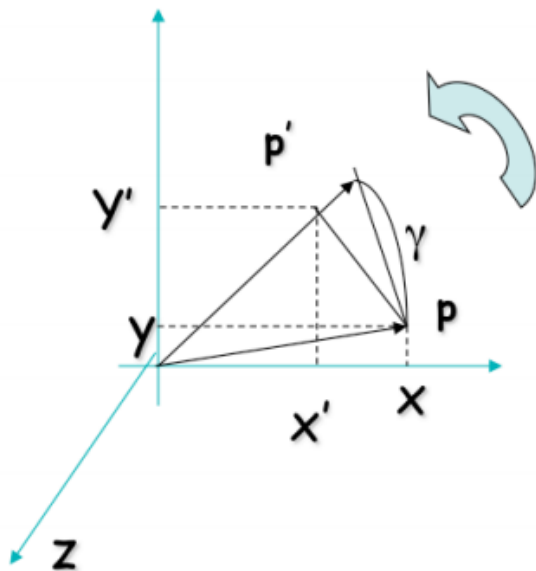


$$T = \begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix}$$

$$P' \rightarrow \begin{bmatrix} \mathbf{I} & T \\ 0 & 1 \end{bmatrix}_{4 \times 4} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Review: 3D rotation

Rotation around the
coordinate axes,
counter-clockwise:



$$R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{bmatrix}$$

$$R_y(\beta) = \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix}$$

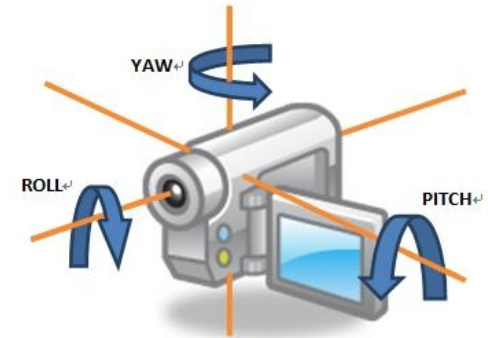
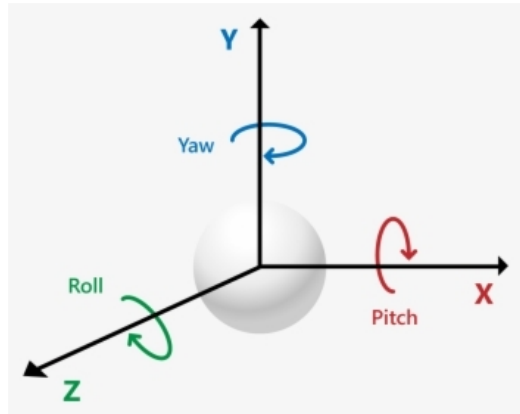
$$R_z(\gamma) = \begin{bmatrix} \cos \gamma & -\sin \gamma & 0 \\ \sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P' \rightarrow \begin{bmatrix} R & 0 \\ 0 & 1 \end{bmatrix}_{4 \times 4} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

A rotation matrix in 3D has 3 degree of freedom

Digression: rotation matrix properties

- Each rotation matrix is orthogonal
 - Proof: try to complement the rotation angle
 - Remember that for an orthogonal matrix $M^{-1}=M^T$
- Given that, the product among orthogonal matrices is still an orthogonal matrix

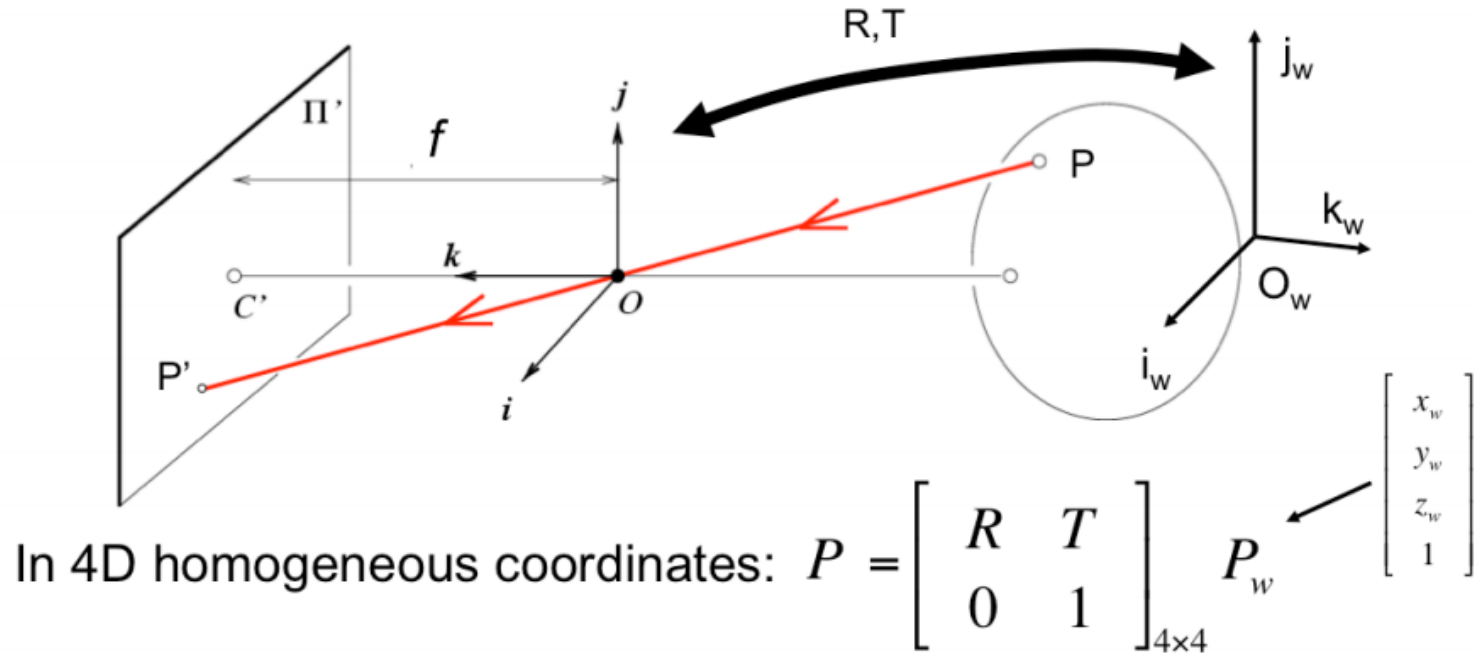


- Rotation order matters!

$$R = R_x(\alpha) R_y(\beta) R_z(\gamma) \quad T = \begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix}$$

$$P' \rightarrow \begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix}_{4 \times 4} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

World Reference System



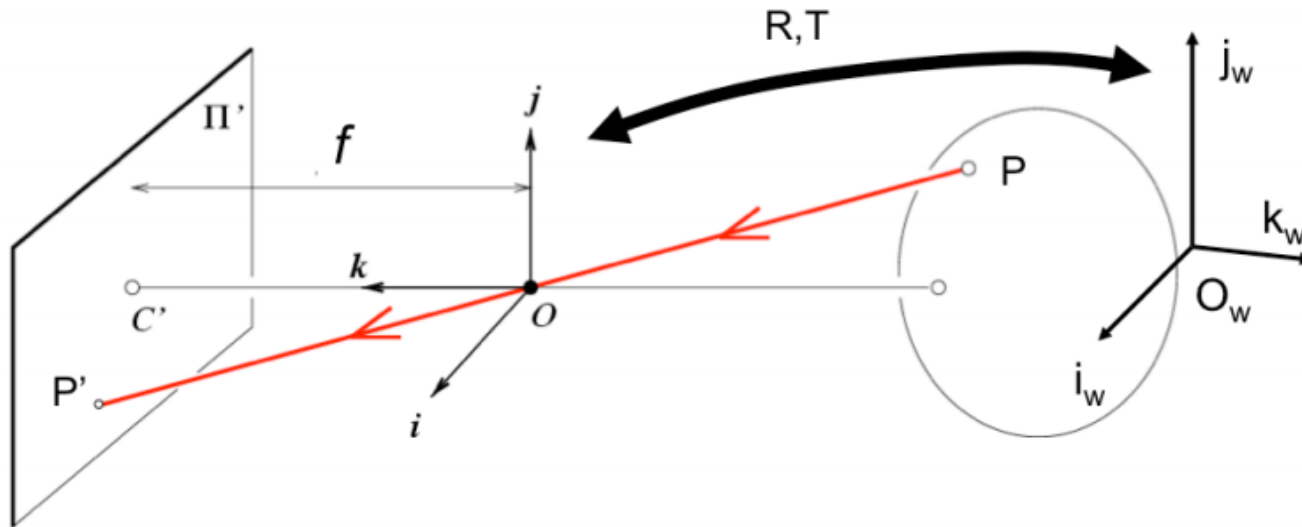
Internal parameters External parameters

$$P' = K \begin{bmatrix} I & 0 \end{bmatrix} P = K \begin{bmatrix} I & 0 \end{bmatrix} \begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix}_{4 \times 4} P_w = \underbrace{K}_{\text{Internal parameters}} \underbrace{\begin{bmatrix} R & T \end{bmatrix}}_{\text{External parameters}} P_w$$

M [Eq.11]

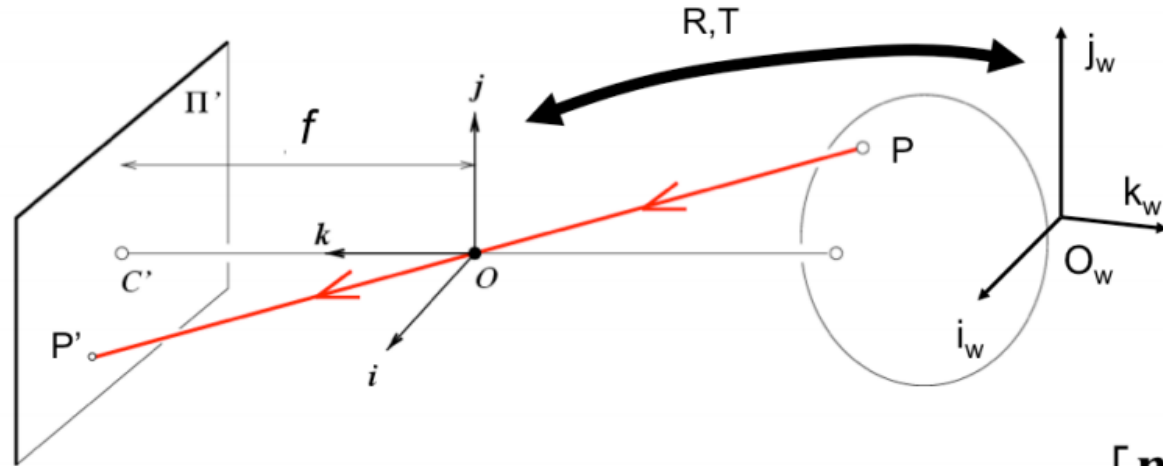
- We added world information
 - Transformation from world reference system to the camera one
 - Extrinsic parameters of a camera only depend on its location and orientation
 - If I “move” the camera they need to be computed again
- $[RT] \rightarrow$ **Camera Extrinsic Matrix**

- 11 degrees of freedom (5 + 3 + 3)



$$P'_{3 \times 1} = M_{3 \times 4} P_w = K_{3 \times 3} \begin{bmatrix} R & T \end{bmatrix}_{3 \times 4} P_{w4 \times 1}$$

Back to Euclidean Coordinates



$$P'_{3 \times 1} = M P_w = K_{3 \times 3} \begin{bmatrix} R & T \end{bmatrix}_{3 \times 4} P_{w4 \times 1} \quad M = \begin{bmatrix} \mathbf{m}_1 \\ \mathbf{m}_2 \\ \mathbf{m}_3 \end{bmatrix}$$

$$= \begin{bmatrix} \mathbf{m}_1 \\ \mathbf{m}_2 \\ \mathbf{m}_3 \end{bmatrix} P_w = \begin{bmatrix} \mathbf{m}_1 P_w \\ \mathbf{m}_2 P_w \\ \mathbf{m}_3 P_w \end{bmatrix} \quad \mathbf{E} \rightarrow \left(\frac{\mathbf{m}_1 P_w}{\mathbf{m}_3 P_w}, \frac{\mathbf{m}_2 P_w}{\mathbf{m}_3 P_w} \right) \quad [\text{Eq.12}]$$

Perspective transformation properties

- Points become.... Points!
- Lines become... Lines!
- Far away objects are smaller (divide by z)



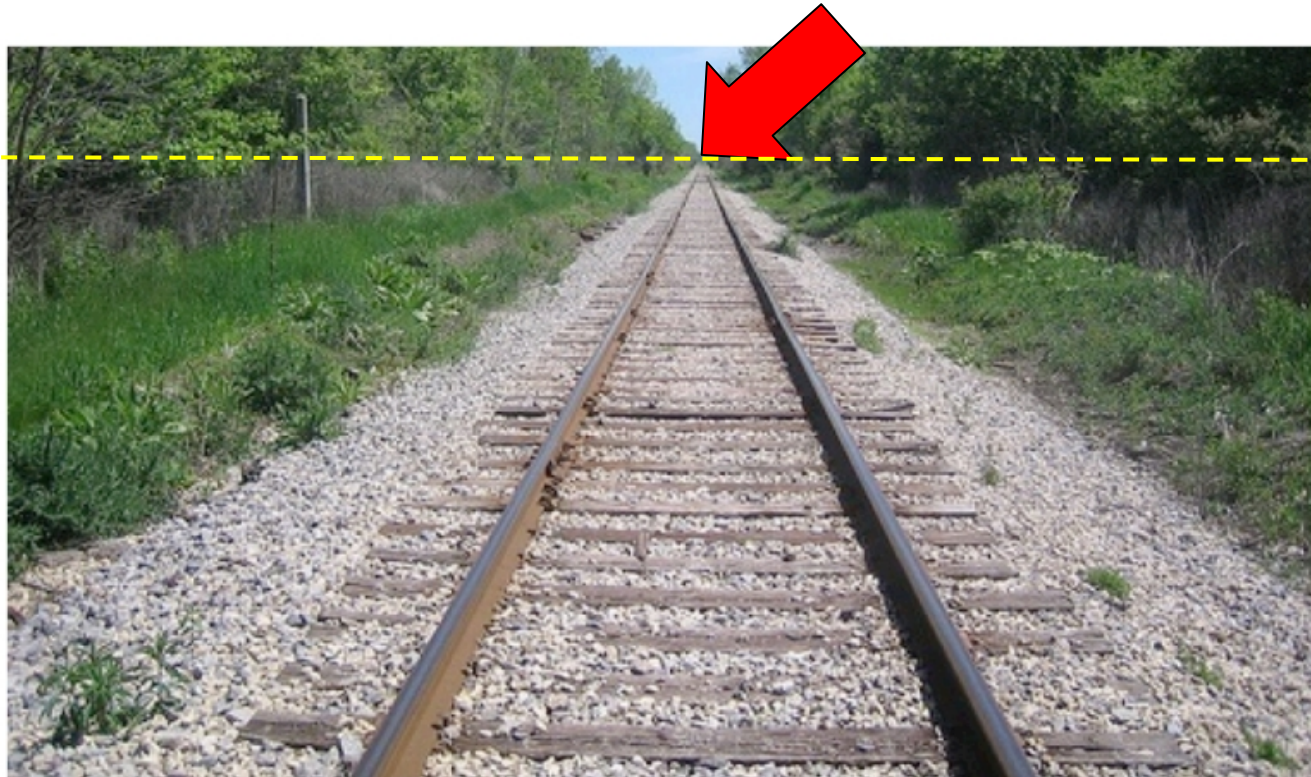
Perspective transformation properties

- Angles are not preserved
- Parallel lines intersect!
 - In the so-called **vanishing point**



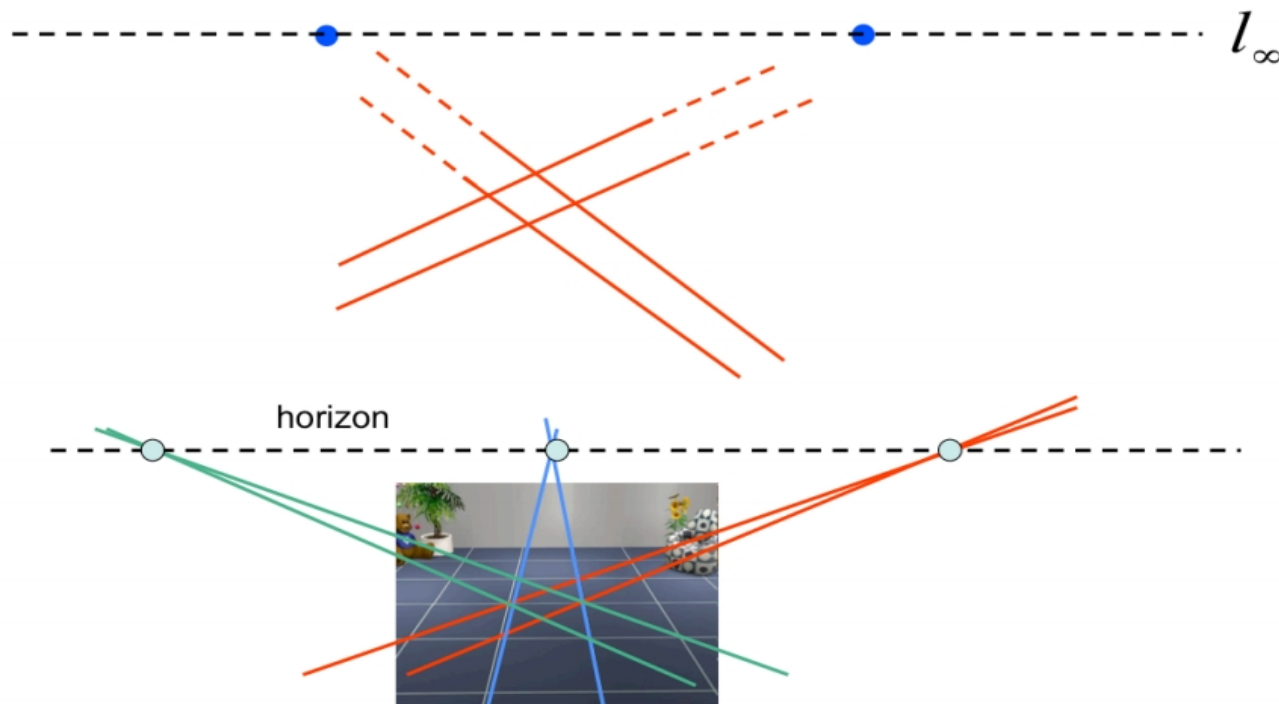
Perspective transformation properties

- Parallel lines that lie in the same plane have vanishing points on a line
- The Horizon (Vanishing Line)



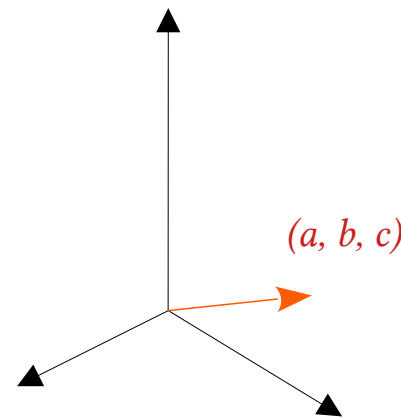
Perspective transformation properties

- Parallel lines that lie in the same plane have vanishing points on a line
- The Horizon (Vanishing Line)



- A 3D straight line can be expressed as:

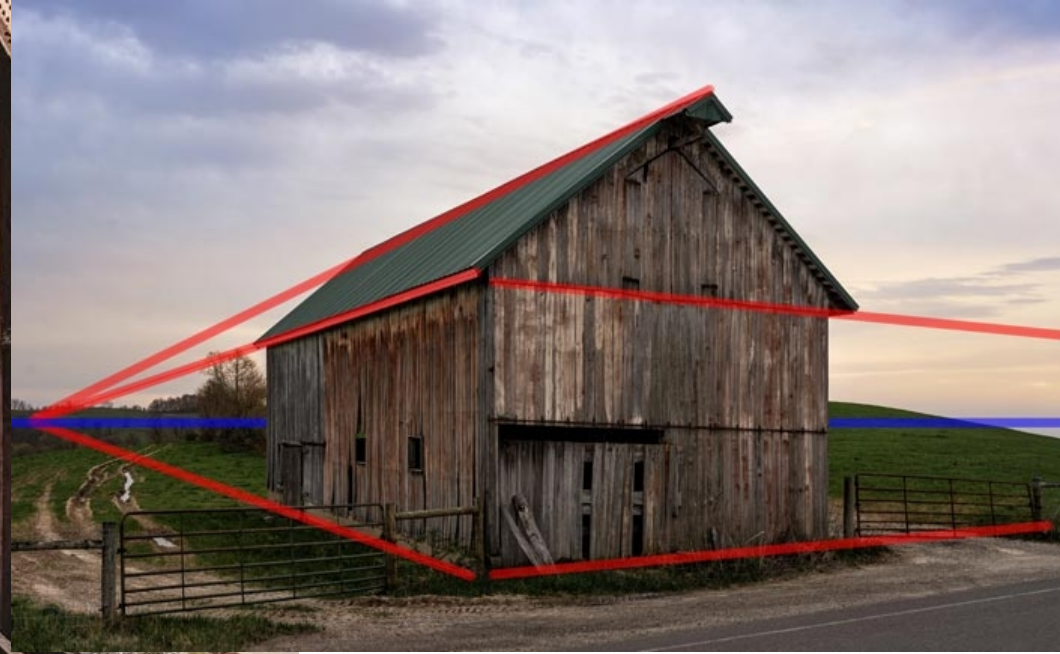
$$\begin{cases} x(t) = x_0 + at \\ y(t) = y_0 + bt \\ z(t) = z_0 + ct \end{cases} \quad \longrightarrow \quad \begin{cases} x'(t) = f \frac{(x_0 + at)}{(z_0 + ct)} \\ y'(t) = f \frac{(y_0 + bt)}{(z_0 + ct)} \end{cases}$$



$$\begin{cases} \lim_{t \rightarrow \pm \infty} x'(t) = f \frac{a}{c} \\ \lim_{t \rightarrow \pm \infty} y'(t) = f \frac{b}{c} \end{cases}$$

- x_0, y_0, z_0 represent a point of the line, where t is the distance from that point.
- a, b, c then encode the direction
- Parallel lines have the same a, b , and c params!
- Points at infinity are not at infinity in the image

Vanishing lines



Pin Hole Geometry Recap

$$M = K \cdot \begin{bmatrix} I & 0 \end{bmatrix} \cdot E = \overset{\text{Intrinsic}}{\boxed{K}} \cdot \underbrace{\begin{bmatrix} R & T \end{bmatrix}}_{\text{Extrinsic}} \in R^{3 \times 4}$$

$$P' = M \cdot P_w = K \cdot \begin{bmatrix} R & T \end{bmatrix} \cdot P_w$$



2D Homogeneous

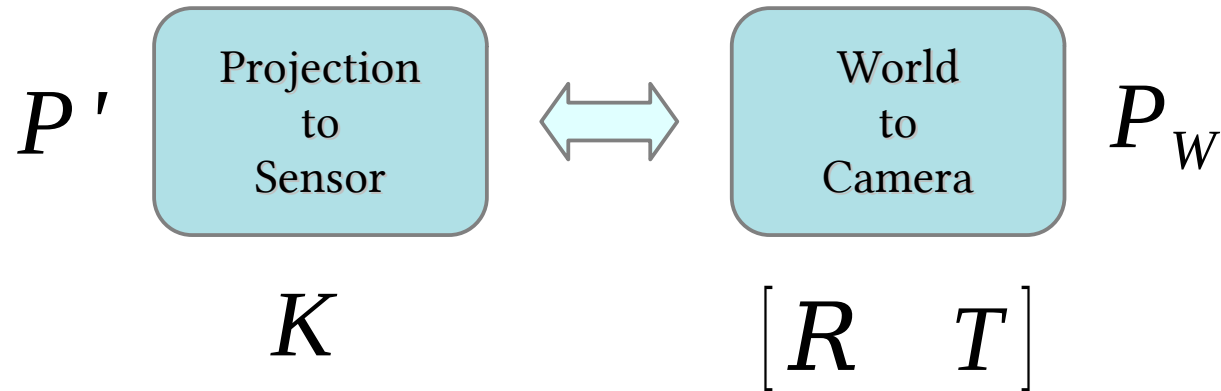


Model the perspective
transformation from 3D to 2D



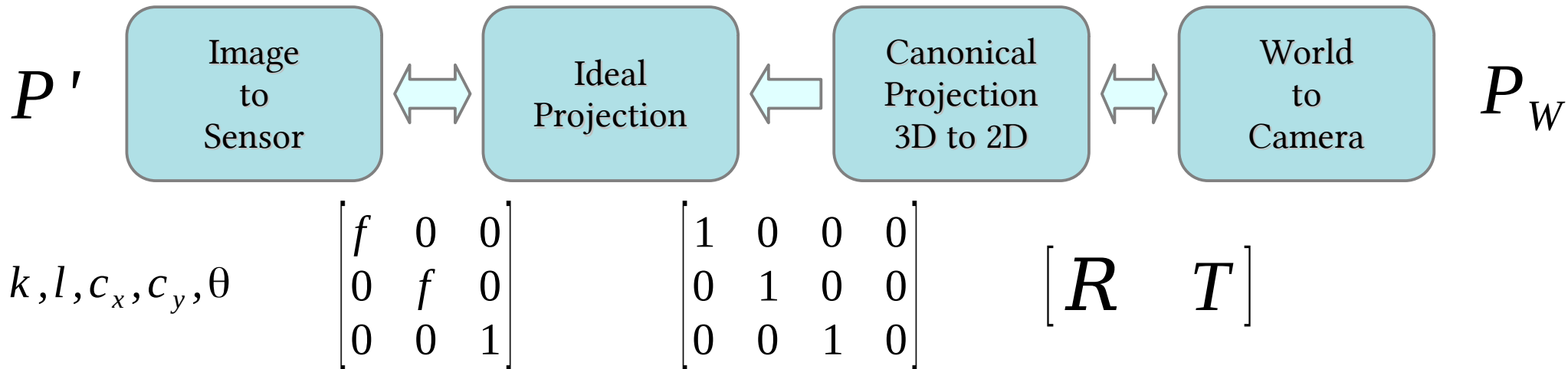
3D Homogeneous

- We can see this transformation as the combination of
 - World to camera
 - Projection to sensor



- The process can be further decomposed
- Also image to sensor can be further decomposed
 - Euclidean, affine, and general

Arrows have a meaning!





UNIVERSITÀ DI PARMA

Camera Models



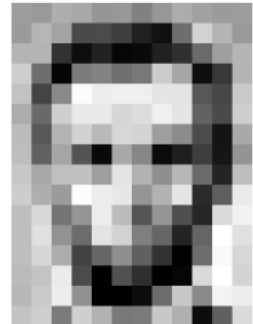
CAMERA MODELS



- Pin Hole Camera
- Lenses
- Pin-Hole Camera Geometry

Courtesy of CS231A · *Computer Vision: from 3D reconstruction to recognition*, Prof. Silvio Savarese – Stanford University

- Since now we discussed image processing
- Namely, we saw fundamental techniques to process a 2D matrix...
- How that image is created?
- What is the relation, if any, to the 3D world?
- During this lesson we will try to answer to those questions

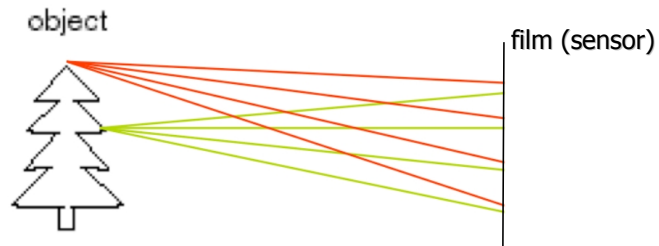


Questa lezione si focalizza su come una immagine viene creata. È importante in quanto la relazione mondo tridimensionale – piano immagine è la base della visione artificiale.

Come detto in origine c'è un mondo dietro all'immagine, e per fare visione, non solo elaborazione immagini occorre capire quale sia la relazione



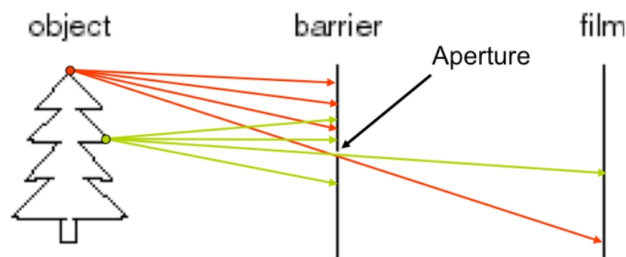
- Too much simple camera
 - A sensitive film in front of an object
 - What is the result?



Mettere semplicemente un sensore “all’aria aperta” non funziona
La luce (solare o artificiale) che colpisce un oggetto, come l’alberello
nell’esempio, viene riflessa in onde circolari (salvo casi strani).
Ad esempio la punta dell’albero riflette tanti raggi e questi si “spalmano” in
maniera più o meno uniforme sul sensore.
Il risultato è inintelligibile



- We can add a barrier with a very small hole
 - So-called pin-hole or aperture
 - Now, only one ray hits the film in a given position



The simplest camera system can be designed by placing a barrier with a small aperture between the 3D object and a photographic film or sensor (that is, a material that is sensitive to light and can be used to record the light that impresses the film/sensor).

As the slide shows, each point on the 3D object emits multiple rays of light outwards; different points and related rays of light are shown in different colors. Only one (or a few) of these rays of light passes through the aperture and hit the film.

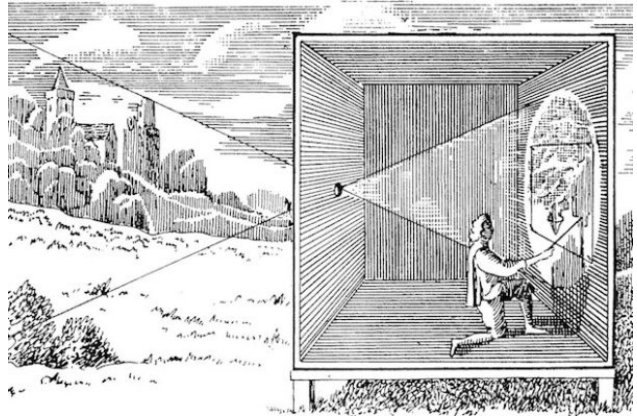
So an approximated 1-to-1 mapping can be established between the 3D object and the film.

The result is that the film gets exposed by an “image” of the 3D object by means of this mapping. This simple camera model is called pinhole camera.



Nel castello di Fontanellato si può visitare una camera oscura.
In realtà un po' più complesso di quanto indicato, c'è un gioco di specchi per
raddrizzare l'immagine

- Milestones
 - Leonardo da Vinci's *Camera Obscura* (1502)

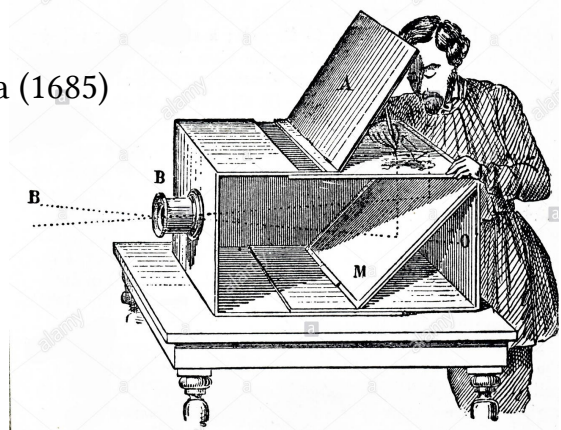


La camera oscura, anche detta camera obscura, camera ottica o fotocamera stenopeica, è un dispositivo ottico composto da una scatola oscurata con un foro stenopeico sul fronte e un piano di proiezione dell'immagine sul retro.
Non è stata inventata ieri



- Milestones

- Leonardo da Vinci's *Camera Obscura* (1502)
- Johan Zahn: first portable camera (1685)



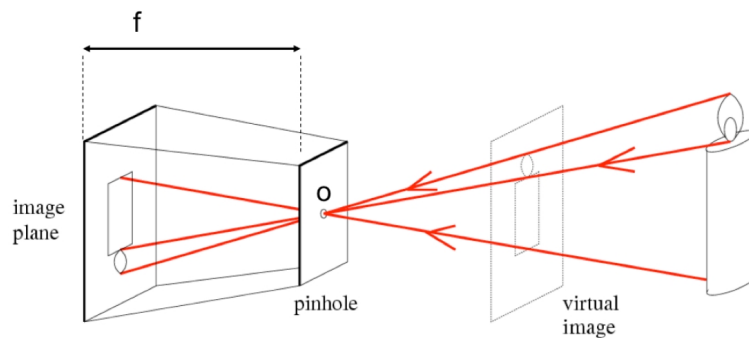
- Milestones

- Leonardo da Vinci's *Camera Obscura* (1502)
- Johan Zahn: first portable camera (1685)
- Joseph Nicéphore Niépce: first photo (1822)



We have to wait until 1822 for first photo which is credited to Joseph Niepce – this marks the birth of photography!
Eliografia, o pittura del sole

- $f \rightarrow$ focal length
- $o \rightarrow$ pin-hole, aperture (center of the lens)



A more formal construction of the pinhole camera model is shown here. The film (now on the left of the object) is commonly called image or retinal plane.

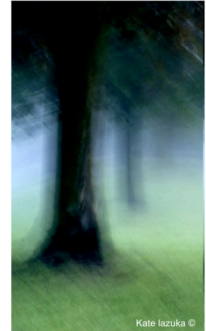
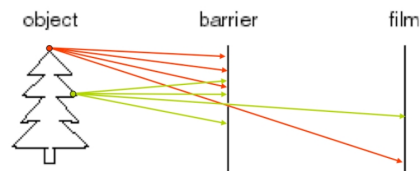
The aperture is called pinhole O or center of the camera.

The distance between the image plane and O is the focal length f . Often, the retinal plane is placed between O and the 3D object at a distance f from O .

In this case it is called virtual image or virtual retinal plane.

Note that the projection of the object in the image plane and the image of the object in virtual image plane are identical up to a scale (similarity) transformation.

- The larger the pin-hole, the greater the number of rays
 - More rays → More light energy
 - More rays → Blurred image



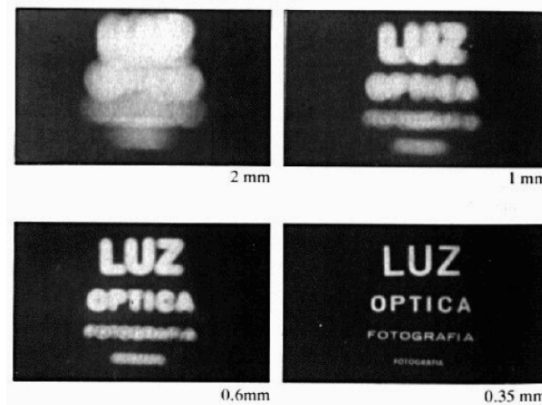
This slide illustrates the importance of the size of the aperture:

The larger is the size of the aperture, the greater is the number of rays of lights that pass through the aperture.

Thus, each point on the film may corresponds to multiple points on the object;
As a result, the image of the object becomes blurred.



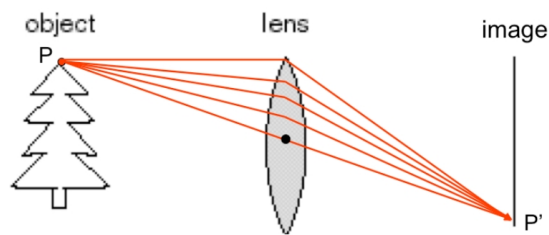
Aperture Size
decrease



On the other hand, the smaller is the size of the aperture, the fewer are the rays of lights that pass through the aperture, which results into having a crispier but darker image of the object. The figure illustrates this property; for a 2 mm aperture the letters look blurred; as the aperture goes from 2 mm to 0.35 mm, the letters look more focused but less bright. Is there a way to overcome this problem? Yes, by adding lenses to replace the aperture!

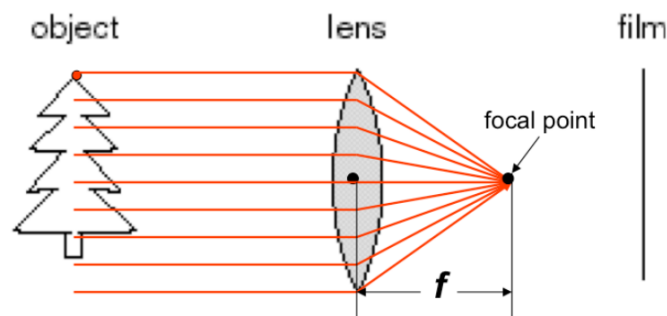


- Light vs Precision
 - Actual pin-holes are barely used
- Lenses are much more comfortable (some issues anyway)
 - We can intercept more rays coming from same 3D point



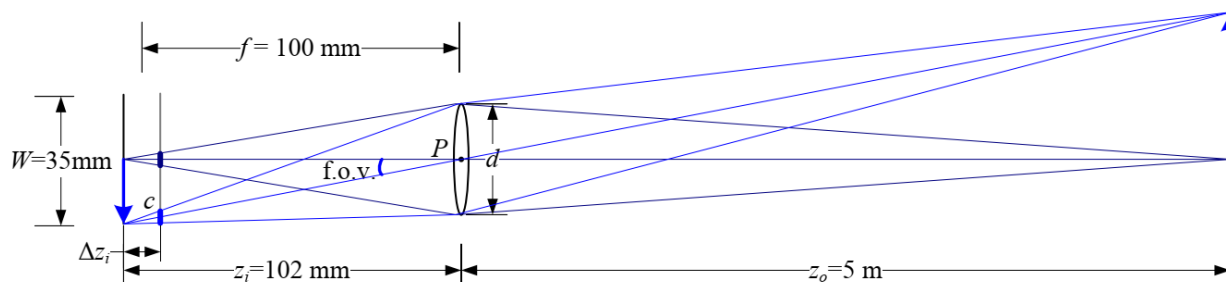
This gives rise to a more complex camera model. If a lens is properly placed and has proper size, it satisfies the following property: all rays of light that are emitted by a point P are refracted by the lens such that they converge to a single point P' in the image plane.

- Parallel light rays converge in a specific point
 - Focal point at distance f from the center of the lens
- Only the ray passing through the center of the lens is not deviated



Lenses have other interesting properties: - A lens focuses light onto the film such that all rays parallel to the optical (or principal) axis converge to one point called focal point. The distance between the focal point and center of the lens is the focal length f . - Rays passing through the center of the lens (i.e., the aperture in the pinhole model) are not deviated

- Only rays emitted from a given plane points in the world converge to points that lie on a specific plane
- The following formula can be used:
$$\frac{1}{f} = \frac{1}{z_i} + \frac{1}{z_o}$$
 - Varying z_o means that also z_i is modified

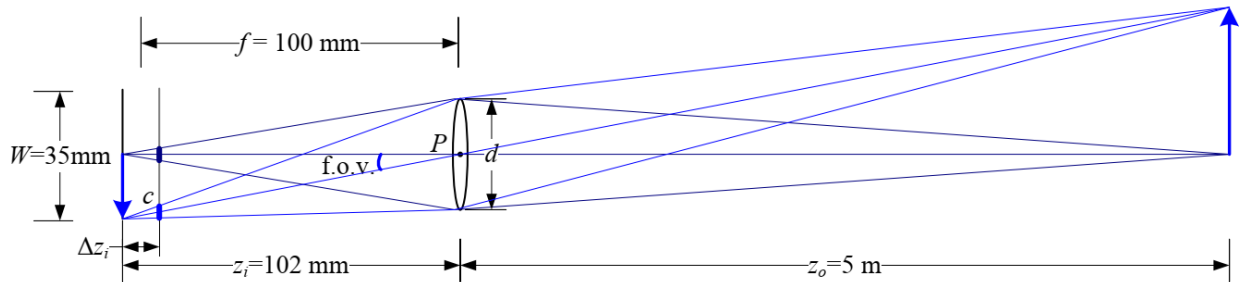


La formula dimostra come solo per un z_o preciso il fuoco è su un dato piano
 Però i pixel hanno una dimensione non nulla $\rightarrow$ profondità di fuoco

A thin lens of focal length f focuses the light from a plane a distance z_o in front of the lens at a distance z_i behind the lens, where $1/z_o + 1/z_i = 1/f$. If the focal plane (vertical gray line next to c) is moved forward, the images are no longer in focus and the circle of confusion c (small thick line segments) depends on the distance of the image plane motion Δz_i relative to the lens aperture diameter d . The field of view (f.o.v.) depends on the ratio between the sensor width W and the focal length f (or, more precisely, the focusing distance z_i , which is usually quite close to f).

Attenzione che se sposto z_0 molto lontano di fatto z_i coincide con f

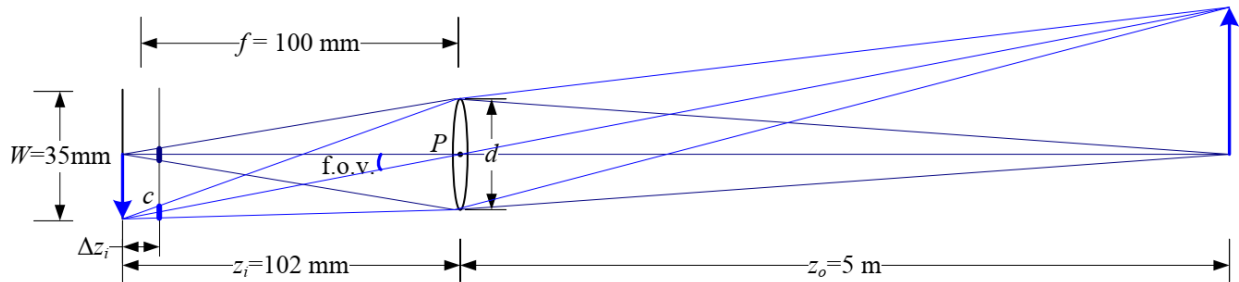
- Anyway in the real world sensor elements have a finite size
- Then we can consider a small portion of the world to be sufficiently in focus: **Shallow Depth of Field**
- Given the $1/z_0$ for z_0 sufficiently large this can mean up to ∞



Il fatto che ogni pixel corrisponda ad un'area e non a un punto infinitesimo fa sì che in realtà non solo un preciso piano sia a fuoco ma un intervallo (che per quanto detto prima potrebbe anche andare all'infinito)
Si può dimostrare che dipende dall'apertura del diaframma

- The depth of field depends on the f-number or focal ratio (f/#)
- The higher the f-number, the larger the depth of field

$$f/\# = \frac{f}{d}$$



Tradizionalmente sarebbe il rapporto tra f e apertura.

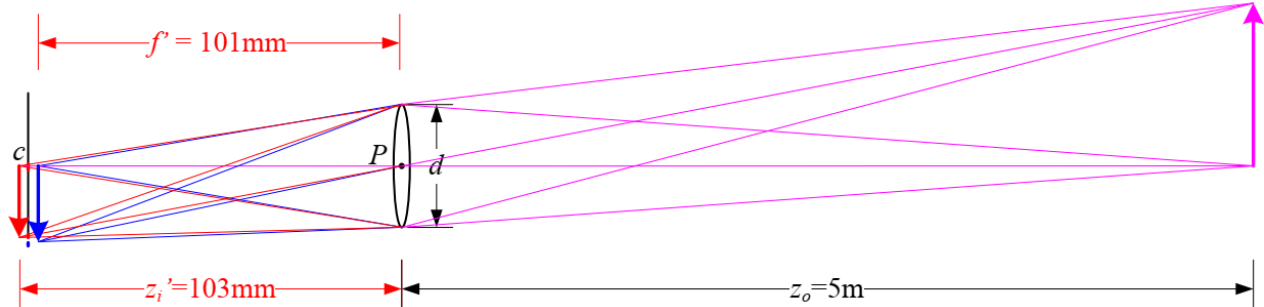


- Example of shallow depth of field



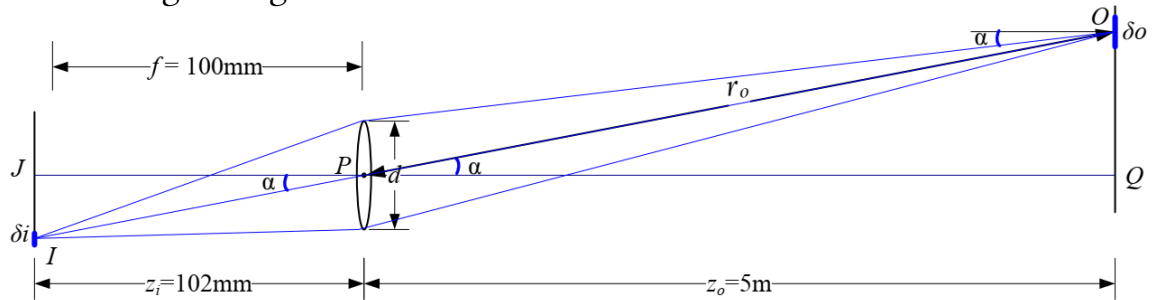
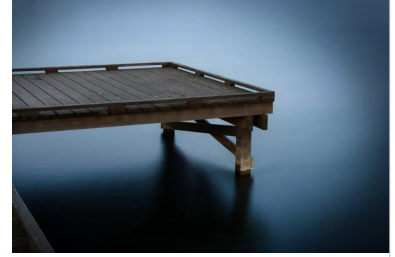
This is true for all the points on the object that are equidistant from the image plane. This no longer true for points that are closer or further to the image plane than P is (e.g. the green point in the figure). The corresponding projection into the image is now blurred or out of focus.

- Refractive index depends on wavelength
- Different colors are then projected in different positions
- **Chromatic Aberration**



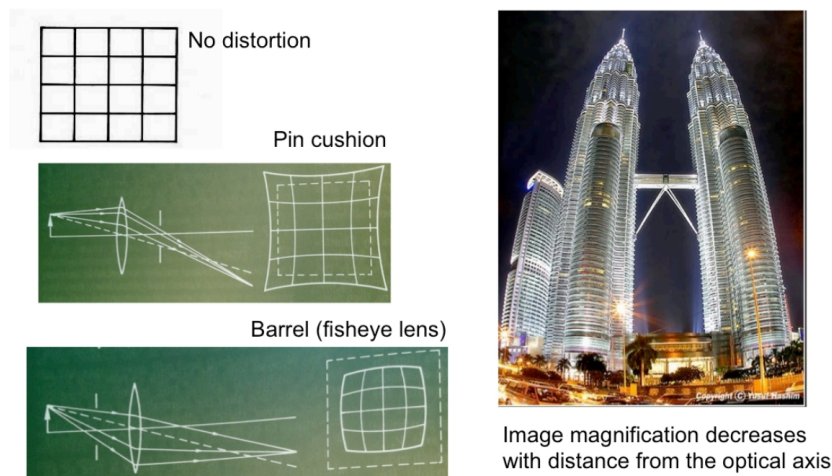
Quanto visto nelle slide precedenti si basa su una ben precisa frequenza. Ma lo spettro del visibile è un intervallo e frequenze diverse portano a rifrazioni differenti, nel caso del visibile 400-700 nm ovvero 430-750 THz
Quindi colori differenti hanno pure distanze di fuoco differenti...

- Areas in the borders/corners are typically darker...
- Distance from optical axis affects energy
 - Vignetting



Vignettatura in pratica nei fatti si riduce l'apertura e quindi quanta luce entra tanto più il punto è lontano dall'asse ottico

Di solito I preprocessing nelle camere lo compensano (per questo non lo vediamo)



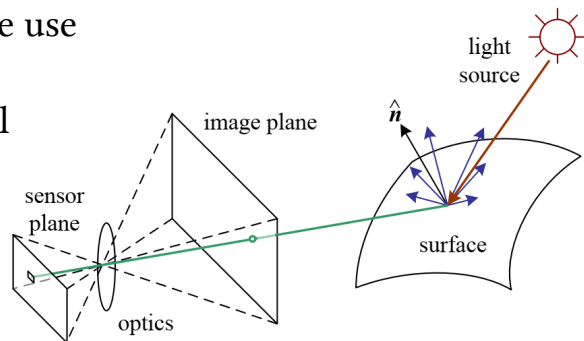
- Distortion effect is much more evident in lateral areas

Because the paraxial refraction model approximates using the thin lens assumption, a number of aberrations can occur. The most common one is referred to as radial distortion, which causes the image magnification to decrease or increase as a function of the distance to the optical axis. We classify the radial distortion as pincushion distortion when the magnification increases and barrel distortion when the magnification decreases.

Radial distortion is caused by the fact that **different portions of the lens have differing focal lengths**.

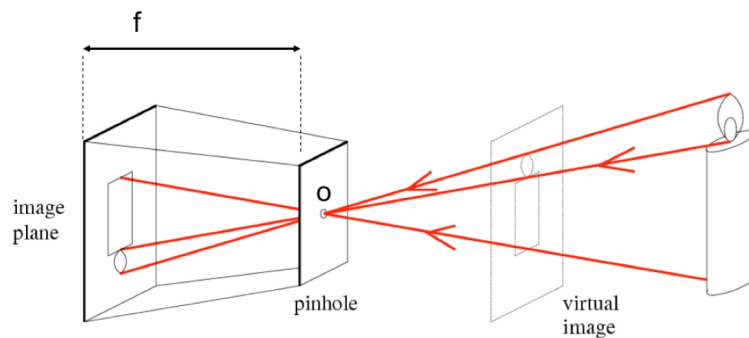
The effect of a radial distortion is that the image magnification decreases or increases as function of the distance separating the optical axis from the point of interest p' . If the magnification increases we talk about pin cushion distortion; if the magnification decreases we talk about barrel distortion. The latter typically occurs when one uses fisheye lens.

- We saw many issues related to the use of a lens
- This also affects the camera model
- Anyway in the following we will simply assume to have:
 - Thin lenses
 - Small angles of view
 - No or compensated chromatic aberration
 - No or rectified distortion effects



Quindi le lenti non sono esattamente equivalenti alla camera oscura MA in alcune specifiche condizioni possiamo ritenerle una buona approssimazione

- $f \rightarrow$ focal length
- $o \rightarrow$ pin-hole, aperture (center of the lens)



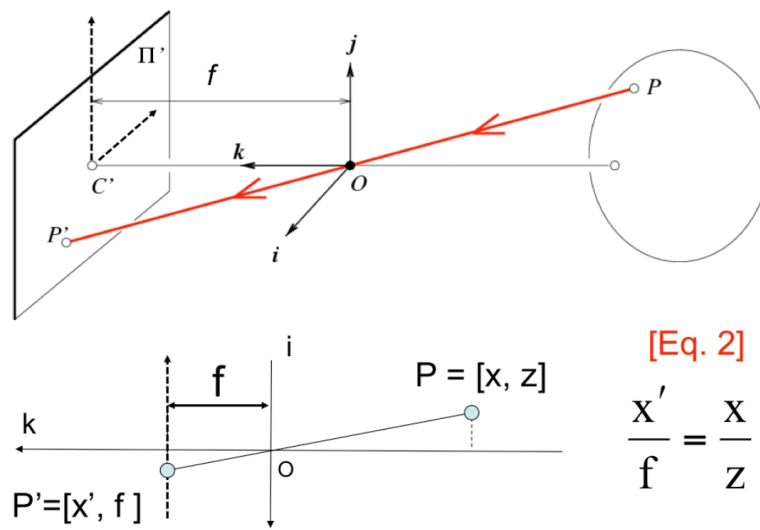
A more formal construction of the pinhole camera model is shown here. The film is commonly called image or retinal plane.

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In this case it is called virtual image or virtual retinal plane.

Note that the projection of the object in the image plane and the image of the object in virtual image plane are identical up to a scale (similarity) transformation.



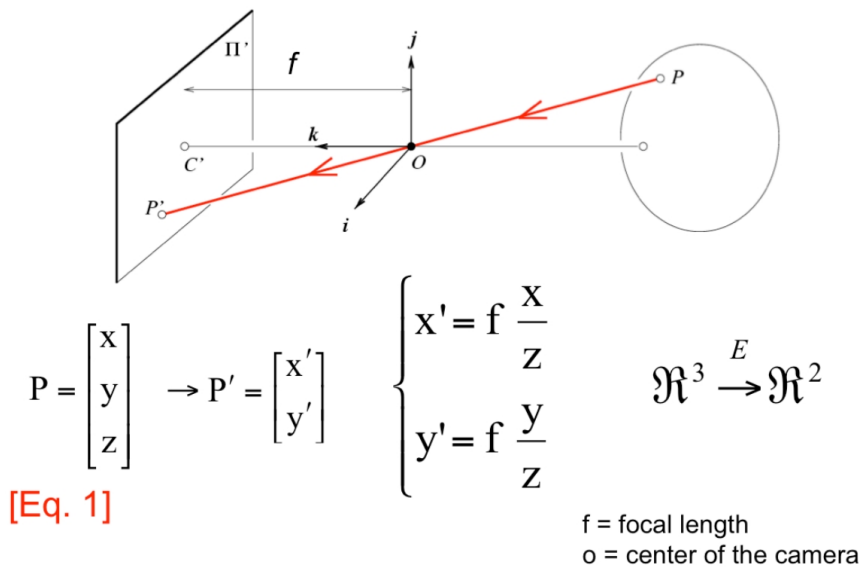
Courtesy: Silvio Savarese

Se ci spostiamo, per semplicità, nel piano 2D I due triangoli $x'fo$ e xzo sono simili e quindi applico le regole dei triangoli simili da cui si evince la eg. 2

Che ovviamente vale anche per la y

We derive the relationship between P and P' for the simpler case where P , O and P' belongs to the same plane defined by i and k (which is perpendicular to j). The bolom figure shows such a plane. The more general case is a simple extension of this and we leave it as an exercise. As the figure shows, the triangle defined by $P'=[x',f]$, O and $[0,f]$ is similar to the triangle defined by P , O and $[0,z]$; from this we can easily derive Eq.2

Pin-Hole Camera: perspective transformation



Courtesy: Silvio Savarese

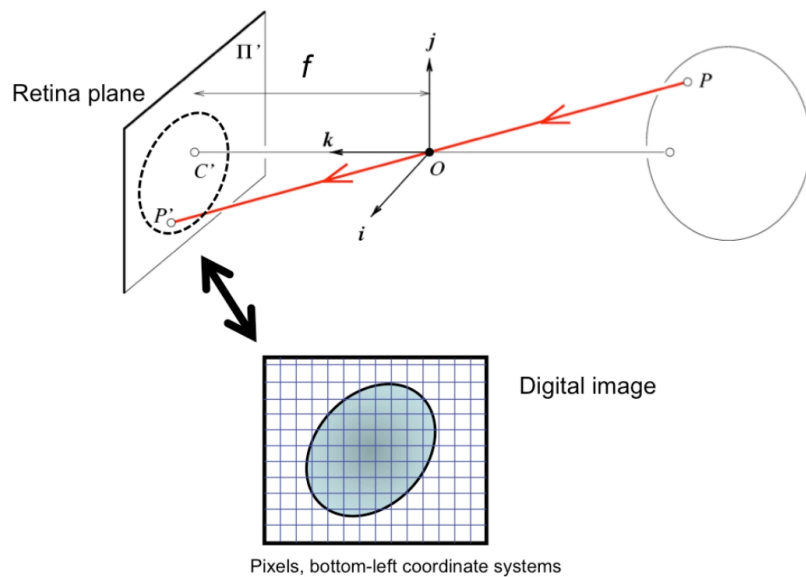
Mettendo tutto insieme si ottiene quel simpatico sistema qui sopra

In the following, we will discuss in details the geometry of the pinhole camera model.

All the results derived using this camera model also hold for the paraxial refraction model.

As we discussed earlier, in the pinhole camera model, a point P in 3D (in the camera reference system) is mapped (projected) into a point P' in the image plane Π' . This is a $\mathbb{R}^3$ to $\mathbb{R}^2$ mapping as indicated by Eq.1. P' can be expressed as $[x' \ y']^T$.

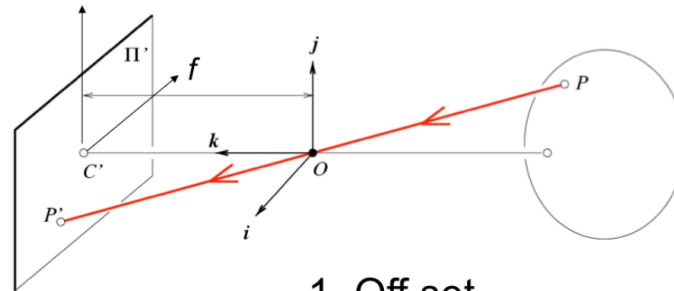
The camera reference system $[i, j, k]$ is centered at O , and the projection of O into the image plane gives C' . This mapping or transformation is also referred to projective transformation.



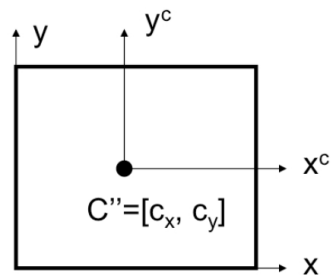
Courtesy: Silvio Savarese

Non basta, infatti vanno considerati altri fattori:

- l'origine nelle immagini la fissiamo in alto a sinistra e non al centro
- le coordinate le diamo in pixel e non in metri
- non è detto che i pixel siano quadrati
- e spesso c'è una distorsione non lineare dovuta alla non perfetta planarità tra sensore e lente



1. Off set

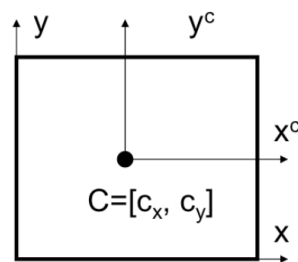
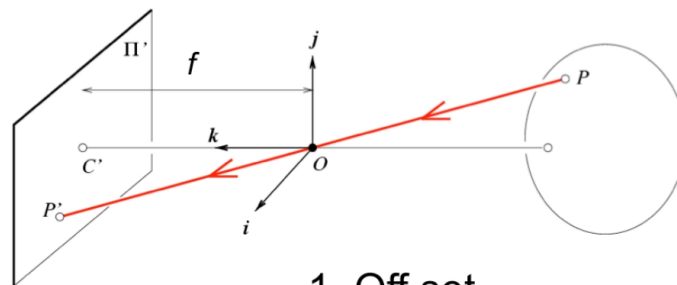


$$(x, y, z) \rightarrow \left(f \frac{x}{z} + c_x, f \frac{y}{z} + c_y \right)$$

[Eq. 5]

Courtesy: Silvio Savarese

Il cambio di origine è semplice, una banale traslazione di C_x e C_y



1. Off set
2. From metric to pixels

$$(x, y, z) \rightarrow \left(\underbrace{f k}_{\alpha} \frac{x}{z} + c_x, \underbrace{f l}_{\beta} \frac{y}{z} + c_y \right) \quad [\text{Eq. 6}]$$

Units: k, l : pixel/m

f : m

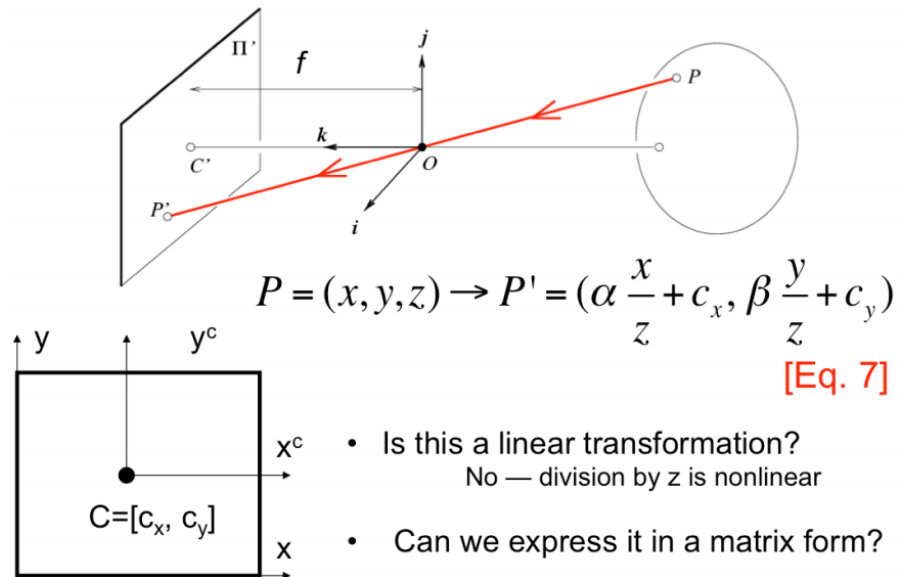
Non-square pixels

α, β : pixel

Courtesy: Silvio Savarese

Il passaggio dal sistema metrico alle coordinate in pixel lo si fa moltiplicando per opportuno fattore di scala o meglio 2 fattori uno che mi dice quanto è largo un pixel in metri e uno quanto è alto. Per semplicità poi usiamo due costanti alfa e beta che sono la moltiplicazione di questi fattori per f (pure lui costante eh?)

I fattori sono diversi in quanto è la regola che i singoli elementi del sensore non siano quadrati



Courtesy: Silvio Savarese

Ci piacerebbe tanto esprimerla in formato matriciale. Lo si potrebbe fare se fosse una espressione lineare.

Ma con uno z al denominatore purtroppo non lo è...

- The non linearity can be solved using Homogeneous Coordinates
- HC are an augmented representation of points
- We add another “coordinate”, i.e. $\mathbb{R}^n \rightarrow \mathbb{R}^{n+1}$
- In 2D space $P=(x, y)$ can be represented as $P=(x, y, 1)$
 - Or more generally as (kx, ky, k)
 - The third value can be considered as a scale factor

C'è però un trucchetto. Fanculo Cartesio e usiamo le coordinate omogenee dette anche coordinate proiettive.

Sì ma cosa sono? In pratica date le coordinate cartesiane in $\mathbb{R}^n$ io aggiungo una ulteriore coordinata (non è esattamente una ulteriore dimensione ma è un fattore di scala). E che valore le assegno? $\rightarrow 1$

Quindi nello spazio 2D passo da (x,y) a $(x,y,1)$

Attenzione però che $(x,y,1)$ è lo stesso punto di coordinate (kx,ky,k) . In pratica a un punto nel piano cartesiano 2D corrispondono infinite possibilità di coordinate omogenee equivalenti.

Dato un generico punto di coordinate omogenee (a,b,c) come torno alla corrispondente coordinata cartesiana 2D? Divido a e b per c ovvero (a,b,c) omogenea diventa $(a/c,b/c)$ cartesiana

- Conversions are simple:
- Euclidean $\rightarrow$ Homogeneous

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$(x, y, z) \Rightarrow \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

- Homogeneous $\rightarrow$ Euclidean

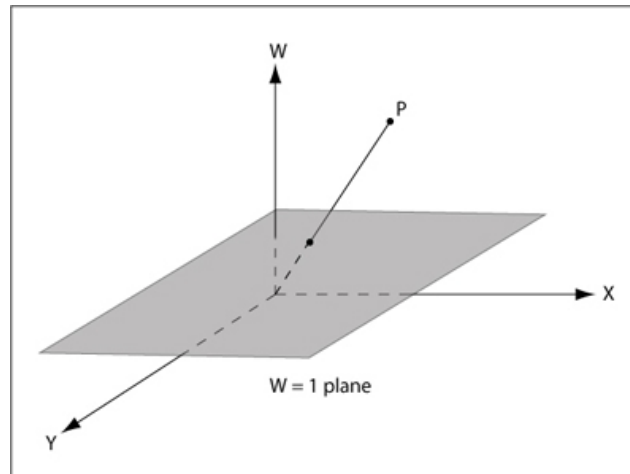
$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w)$$

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \Rightarrow (x/w, y/w, z/w)$$

One way to get around this problem is to change coordinate system and go from the Euclidean reference system (our original coordinate system) to the so called Homogenous coordinate system. The process of going back and forth from Euclidean and Homogenous coordinate system is non-linear, but the good news is that once we are in the homogenous coordinate system a projective transformation like the one expressed by Eq. 7 becomes linear. We will discuss this in a few more details next. Please refer to HZ section 2.2 for more details and examples. The change of reference system from Euclidean to Homogenous goes as follows: E $\rightarrow$ H: a point X in R^n (in the Euclidean coordinate system) is represented as R^{n+1} vector (in the Homogenous coordinate system) by adding a final coordinate of 1. See slide for examples. H $\rightarrow$ E: a generic point X in R^{n+1} (in the Homogenous coordinate system) is represented as a R^n vector (in the Euclidean coordinate system) by suppressing the last coordinate and dividing the first n coordinates of X by the n+1th coordinate. See slide for examples. The benefit of working with homogenous coordinates will become more evident throughout the upcoming lectures. For now let's go back to Eq. 7 and propose an alternative formulation in the homogenous coordinate systems.



- A geometric interpretation for HC can be given as follows



Prendiamo ad esempio il piano 2D. Aggiungere una coordinata all'atto pratico è come passare da 2D a 3D.

Il piano originale è quello per cui $W=1$. Tutte le coordinate omogenee possibili del punto originale giacciono di fatto su una retta che passa dall'origine e dal nostro punto $(x,y,1)$

Perfetto esempio di trasformazione proiettiva!

- Three main reasons to use homogeneous coordinates
 - Simple representation of points and lines (no special cases)
 - Homogenous space contains more points than Euclidean one!
 - $(x,y,0)$
 - Simple representation of Euclidean Transformations
 - Traslation
 - Scale
 - Rotation
 - Simple representation of perspective projections

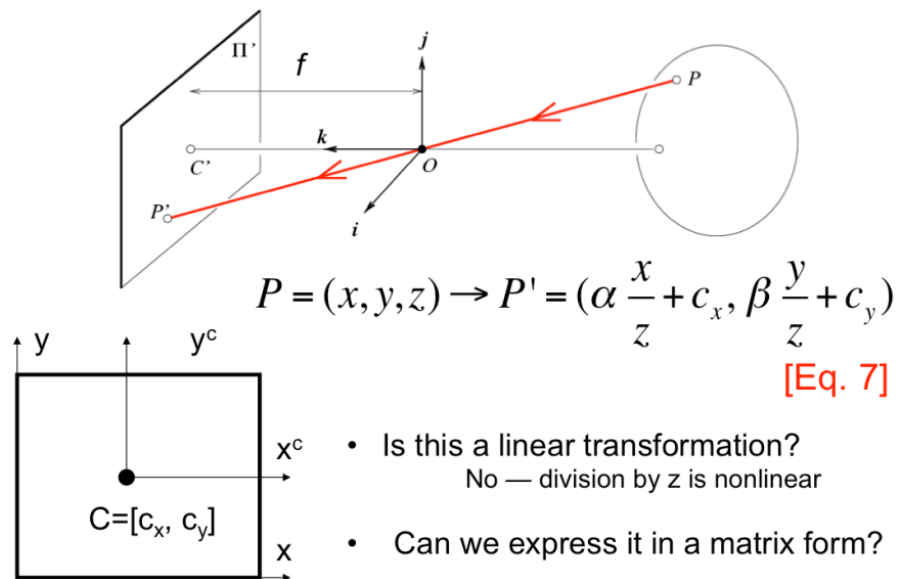
I punti all'infinito sono detti Ideal Points e giacciono tutti sulla linea ideale $(0,0,1)$

Comunque quello che conta sono che:

Dato un punto (x,y,w) in coordinate omogenee io lo faccio sempre corrispondere ad un punto 2D euclideo (salvo infiniti vari)

Le operazioni che abbiamo visto prima diventano esprimibili in forma matriciale!

Non Linear Transformation (again)



Courtesy: Silvio Savarese

Torniamo alla nostra espressione non lineare e consideriamo però adesso di usare le relative coordinate omogenee.



- $P \rightarrow P'$ projection becomes $P_h \rightarrow P_h'$
- The $P=[x \ y \ z]$ in the 3D space is $P_h=[x \ y \ z \ 1]$ in Homogeneous reference system
- P' was computed as $[\alpha(x/z)+c_x \ \beta(y/z)+c_y]$
- P_h' can be then $[\alpha x+c_x z \ \beta y+c_y z \ z]$
- Do you see how to express $P_h \rightarrow P_h'$ as matrix product?

Per il momento usiamo un h al pedice per differenziare tra coordinate cartesiane e coordinate omogenee (ma poi la faremo sparire)

Per esprimere in coordinate omogenee il mio P nel mondo aggiungo semplicemente un 1 come quarta coordinata

Nel caso del P immagine posso fare lo stesso. Ma vi ricordate che ho infinite possibilità? Basta semplicemente moltiplicare tutte e 3 le coordinate omogenee per un fattore di scala. Quindi aggiungo 1 e moltiplico per z (che va bene come fattore di scala)

Boom! La non linearità sparisce e posso adesso esprimere il tutto in forma matriciale!

$$P_h' = \begin{bmatrix} \alpha x + c_x z \\ \beta y + c_y z \\ z \end{bmatrix} = \begin{bmatrix} \alpha & 0 & c_x & 0 \\ 0 & \beta & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \quad \text{--- } P_h \quad \text{[Eq.8]}$$

Homogenous Euclidian

$$P_h' \rightarrow P' = \left(\alpha \frac{x}{z} + c_x, \beta \frac{y}{z} + c_y \right)$$

$$M = \begin{bmatrix} \alpha & 0 & c_x & 0 \\ 0 & \beta & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

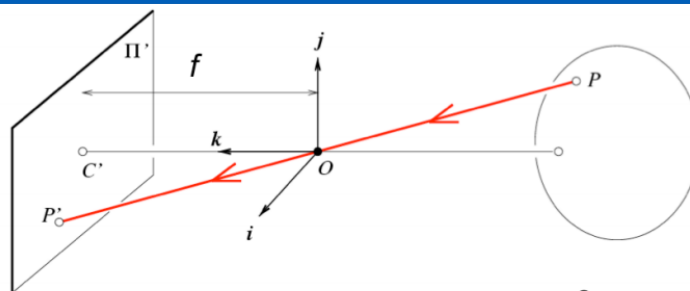
Courtesy: Silvio Savarese

A point $P=[x \ y \ z]^T$ in the 3D Euclidean reference system is expressed as $P_h = [x \ y \ z \ 1]^T$ in Homogenous reference system. Let's multiply the matrix M (which contains an arrangement of the camera parameters) by P_h : the result is the vector $P'h$ as expressed by Eq. 8. Now let's convert $P'h$ back to the Euclidean reference system.

The result is exactly the point P' in Eq. 7. As this process illustrates:

- **The projective transformation can be expressed as a linear transformation in the Homogenous reference system**
- **This transformation is expressed as a multiplication of the matrix M and the vector P_h .**
- The conversion from Homogenous to Euclidean returns the actual point P' (that's where the non-linearity takes place).

The key advantage of this representation is that we have obtained a much more compact expression for capturing the projective transformation. As we'll see next, as we consider additional linear transformations to characterize the camera model, these can be nicely and compactly added as matrix multiplications to the expression in Eq. 8



[Eq.9]

$$P' = M P$$

$$= K \begin{bmatrix} I & 0 \end{bmatrix} P$$

$$P' = \begin{bmatrix} \alpha & 0 & c_x & 0 \\ 0 & \beta & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Camera
matrix K

Courtesy: Silvio Savarese

Nella notazione facciamo sparire l'h pedice. Da ora in poi solo calcolo matriciale e coordinate omogenee.

The intrinsic matrix transforms 3D camera coordinates to 2D homogeneous image coordinates.

La matrice M è una matrice 3x4 la posso vedere come una matrice K 3x3 moltiplicata per una matrice 3x4 di cui i primi 3x3 elementi sono una matrice identità e poi c'è una colonna di 0.

Notazione che useremo spesso

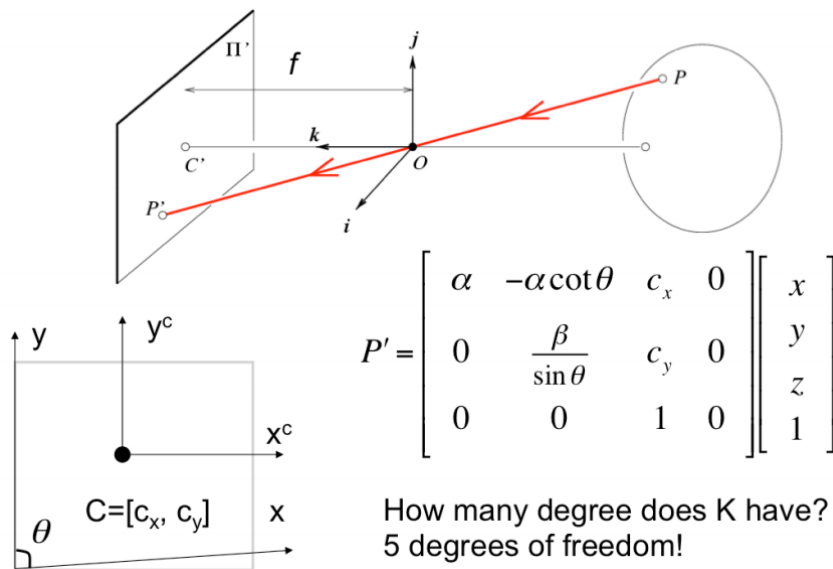
La K è la cosiddetta matrice degli intrinseci e codifica i parametri specifici della telecamera (in realtà nelle slide successive la modifichiamo leggermente per tener conto dello skew, andrebbe anche considerata la distorsione ma passiamo la mano per il momento)

From this point on, let's assume that we work on homogenous coordinates and that a generic point P in 3D is expressed in such coordinate system. To keep the notation light, we drop the index h (we'll add the index back when the discussion becomes ambiguous). By working in homogenous coordinates, we can write a very compact expression that regulates the projective transformation (see Eq. 9), where M is defined in the previous slide, I is a 3x3 identity matrix and K is defined as the camera matrix. This matrix contains some of the critical parameters that are useful to characterize a camera model. Two parameters are missing: skewness and distortion. We will talk about skewness next and distortion during the next lecture.



- Not enough, we have to consider skewness!
- Sometimes, the 2D image plane is not a rectangle but rather is skewed
 - i.e. the angle between the image axis is not 90 degrees.
- Another transformation needs to be carried out to go from the rectangular plane to the skewed plane
- No demonstration

Posso esser bravo e preciso quanto voglio ma non riuscirò mai a installare sensore e lente in maniera precisa. Quindi è possibile che l'asse ottico non sia perpendicolare al sensore. Inoltre anche l'acquisizione introduce una specie di skewness dovuta a sincronizzazione del sensore (non tutti i pixel sono acquisiti allo stesso istante, cosa che peggiora con il rolling shutter sebbene in questo caso non ha neanche senso parlare di skew)



Courtesy: Silvio Savarese

All'atto pratico è come se gli assi x e y non fossero perpendicolari ma avessero tra loro un angolo differente. Questo mi modifica la mia K (ci fidiamo del risultato eh?)

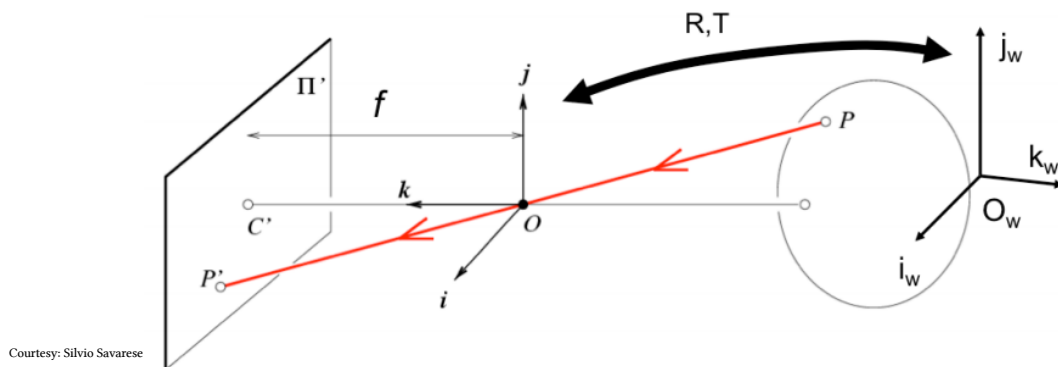
- So far we adopted a camera centric approach
 - Reference system centered on the pin-hole
 - All parameters are camera dependent only
 - No external world
- $K \rightarrow$ **Camera Intrinsic Matrix**

Nelle slide precedenti abbiamo fissato l'origine direttamente sul pinhole.

Un po' limitante. In genere avrò una origine nel mondo (basti pensare a quando uso più di un sensore, dovrò pur usare un sistema di coordinate unico per tutti no?)

Introducing the external world...

- Having a camera reference system is a bit limiting
- Usually a different reference system is used
- We need an additional transformation

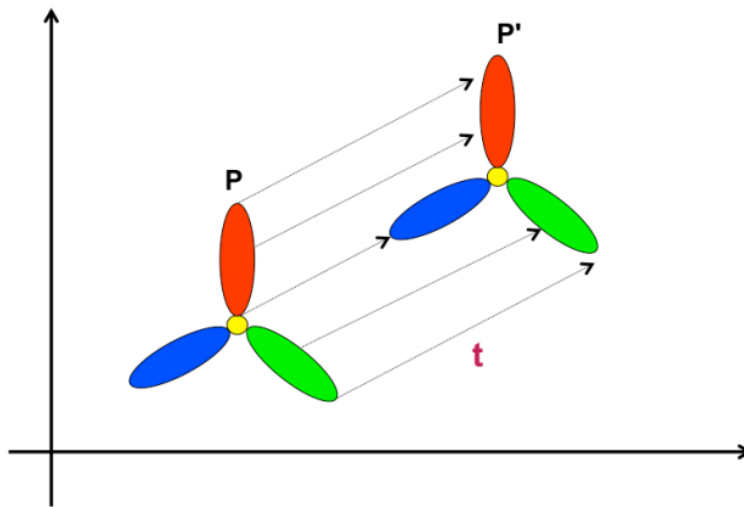


Courtesy: Silvio Savarese

Stiamo parlando del sistema di riferimento di P ovviamente. Per convertire dall'uno all'altro devo usare una rototraslazione

So far we have described a mapping between a point P in the 3D camera reference system to a point P' in the 2D image plane. But what if the information we have about points in 3D (or in general about any physical object) is available in the world coordinate system? Then, we need to include to our mapping an additional transformation that allows to relate points from/to the world reference system to/from the camera reference system. This transformation is captured by a rotation matrix R and a translation vector T . Next we'll review the basics of rigid transformations.

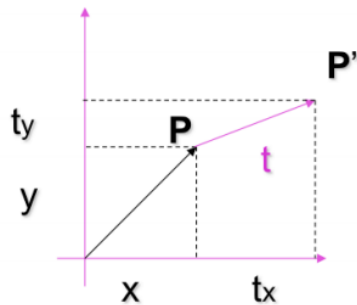
Review: 2D translation



Courtesy: Silvio Savarese

Per semplicità vediamo cosa significa rototraslare in 2D (l'estensione al 3D è semplice)

Review: 2D translation



$$\mathbf{P} = (x, y)$$

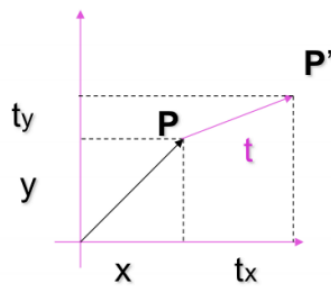
$$\mathbf{t} = (t_x, t_y)$$

$$\mathbf{P}' = \mathbf{P} + \mathbf{t} = (x + t_x, y + t_y)$$

Courtesy: Silvio Savarese

Traslazione: mi sposto di un vettore $\mathbf{t}$ in pratica sommo le componenti t_x e t_y di quel vettore alle precedenti coordinate

Review: homogeneous 2D translation

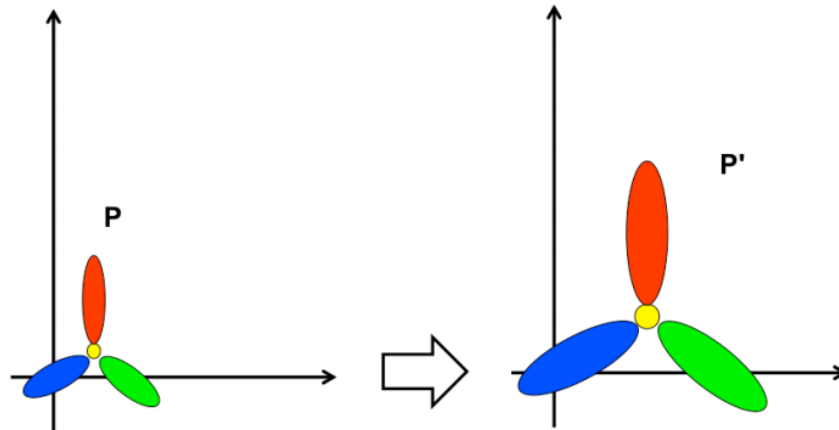


$$\mathbf{P} = (x, y) \rightarrow (x, y, 1)$$

$$\mathbf{P}' \rightarrow \begin{bmatrix} x + t_x \\ y + t_y \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$
$$= \begin{bmatrix} \mathbf{I} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \mathbf{T} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Courtesy: Silvio Savarese

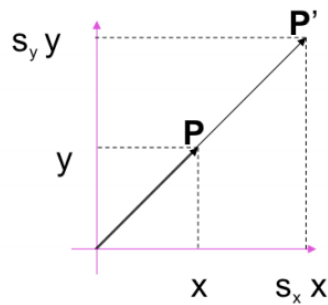
In forma matriciale e usando le coordinate omogenee lo posso scrivere così



Courtesy: Silvio Savarese

La scala non la useremo ma vediamola comunque

Review: homogeneous 2D scaling



$$\mathbf{P} = (x, y) \rightarrow \mathbf{P}' = (s_x x, s_y y)$$

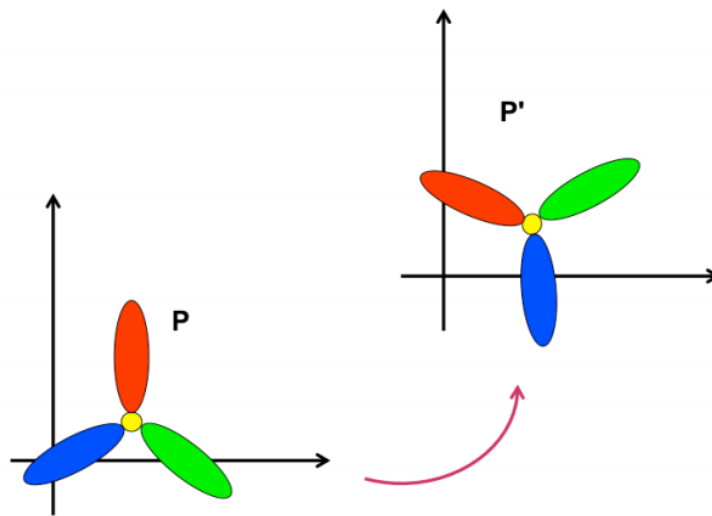
$$\mathbf{P} = (x, y) \rightarrow (x, y, 1)$$

$$\mathbf{P}' \rightarrow \begin{bmatrix} s_x x \\ s_y y \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{S}} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{S}' & \mathbf{0} \\ \mathbf{0} & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \mathbf{S} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Courtesy: Silvio Savarese

In pratica moltiplico le due coordinate per un fattore e anzi posso pensare di usare fattori di scala differenti in orizzontale e verticale. Anche qui è facile metterle in forma matriciale (era fattibile anche in coordinate cartesiane direttamente)

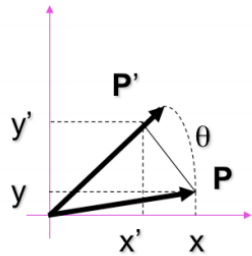
Review: 2D rotation



Courtesy: Silvio Savarese

LA rotazione in 2D è semplice

- Rotate around the z axis by Θ



$$x' = \cos \theta x - \sin \theta y$$

$$y' = \cos \theta y + \sin \theta x$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\mathbf{P}' = \mathbf{R} \mathbf{P}$$

How many degrees of freedom? 1

$$\mathbf{P}' \rightarrow \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Courtesy: Silvio Savarese

Ruoto attorno all'origine e applico una matrice di rotazione già volendo a livello di coordinate cartesiane. L'estensione a quelle omogenee è banale

Review: put everything together



$$\begin{aligned}
 \mathbf{P}' &\rightarrow \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \\
 &= \begin{bmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \\
 &= \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{S} & \mathbf{0} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R} \mathbf{S} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}
 \end{aligned}$$

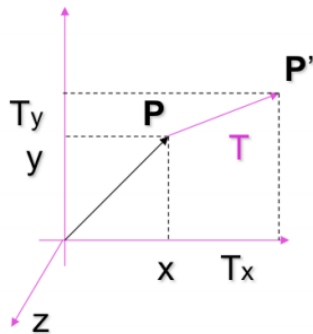
If $s_x = s_y$, this is a similarity transformation

Courtesy: Silvio Savarese

Mettiamo tutto insieme. Basta applicare le trasformazioni in sequenza e ottenere una matrice 3x3 unica. Uso la notazione compatta già vista.

Questa trasformazione è detta di similarità in quanto mantiene la forma degli “oggetti”

Review: 3D translation



$$T = \begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix}$$

$$P' \rightarrow \begin{bmatrix} I & T \\ 0 & 1 \end{bmatrix}_{4 \times 4} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

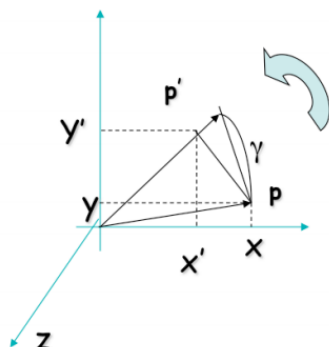
Courtesy: Silvio Savarese

A translation vector in 3D has 3 degrees of freedom

L'estensione al 3D della traslazione è semplice (attenzione è una matrice 4x4)

Review: 3D rotation

Rotation around the
coordinate axes,
counter-clockwise:



$$R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{bmatrix}$$

$$R_y(\beta) = \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix}$$

$$R_z(\gamma) = \begin{bmatrix} \cos \gamma & -\sin \gamma & 0 \\ \sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P' \rightarrow \begin{bmatrix} R & 0 \\ 0 & 1 \end{bmatrix}_{4 \times 4} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Courtesy: Silvio Savarese

A rotation matrix in 3D has 3 degree of freedom

La rotazione no, è un pasticcio. Oltretutto ci sono differenti sistemi per indicare gli angoli di rotazione. Il più diffuso è il sistema di Eulero. Ve lo lascio descritto qui sotto per capire la complicazione.

Resta il fatto che se moltiplico quelle 3 rotazioni tra di loro ottendo comunque una matrice 3x3 per il piano cartesiano che devo estendere per le coordinate omogenee a 4x4

The Euler angles are three angles introduced by Leonhard Euler to describe the orientation of a rigid body with respect to a fixed coordinate system

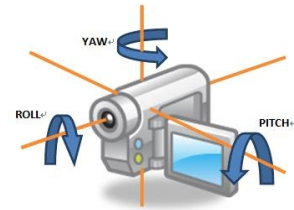
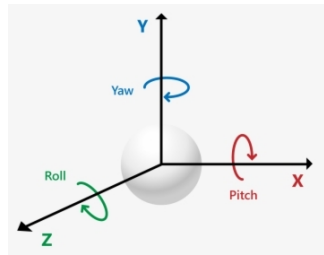
N axis is the intersection between xy plane and new xy plane

Alpha angle between x axis and N axis

Beta is the angle between the 2 z axis

Gamma is the axis between new x axis and N axis

- Each rotation matrix is orthogonal
 - Proof: try to complement the rotation angle
 - Remember that for an orthogonal matrix $M^{-1}=M^T$
- Given that, the product among orthogonal matrices is still an orthogonal matrix



Una matrice ortogonale è tale che la trasposta è uguale all'inversa e in questo caso invertire la matrice è uguale ad invertire la rotazione

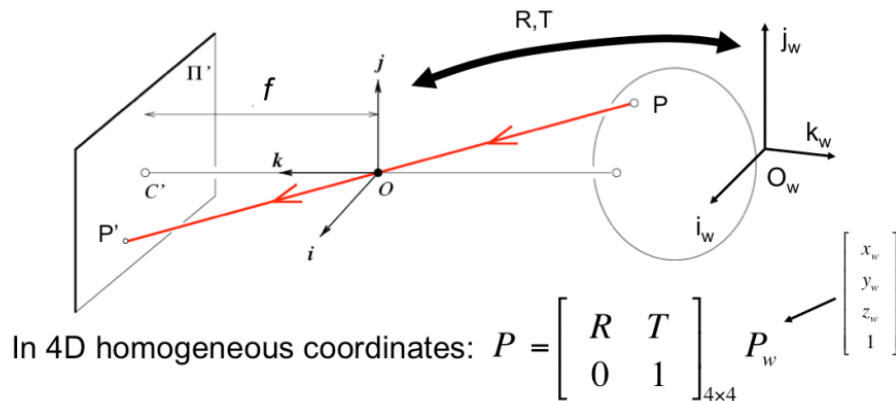
- Rotation order matters!

$$R = R_x(\alpha) R_y(\beta) R_z(\gamma) \quad T = \begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix}$$

$$P' \rightarrow \begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix}_{4 \times 4} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Courtesy: Silvio Savarese

The Euler angles are three angles introduced by Leonhard Euler to describe the orientation of a rigid body with respect to a fixed coordinate system



$$P' = K \begin{bmatrix} I & 0 \end{bmatrix} P = K \begin{bmatrix} I & 0 \end{bmatrix} \begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix}_{4 \times 4} P_w = \underbrace{K}_{\text{Internal parameters}} \underbrace{\begin{bmatrix} R & T \end{bmatrix}}_{\text{External parameters}} P_w$$

M [Eq. 11]

Courtesy: Silvio Savarese

Mettiamo tutto insieme!

The transformation that allows to relate points from/to the world reference system to/ from the camera reference system is a rigid transformation and is regulated by a translation R and a rotation T in 3D. The benefit of working in a homogeneous coordinates is that these two transformations can be compactly described by the matrix $[R \ T; \ 0 \ 1]$ as the slide illustrates.

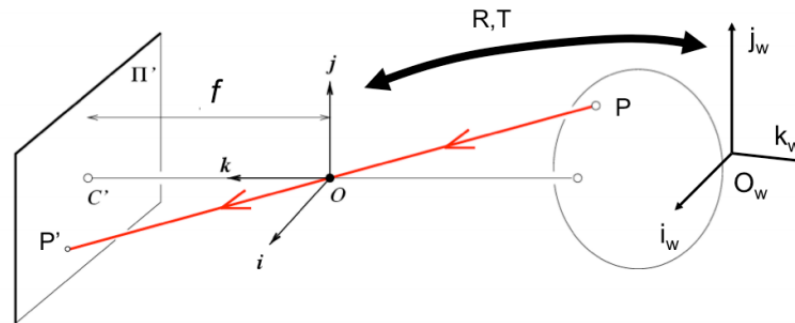
This is a 4x4 matrix and allows to map any point $P_w = [x_w \ y_w \ z_w \ 1]^T$ in the world reference system to a point $P = [x \ y \ z \ 1]^T$ in the camera reference system (in homogenous coordinates).

The projective mapping from P_w to $P' = [x' \ y' \ z']$ is expressed by Eq. 11 and is compactly summarized as $P' = M P$. The projection matrix $M = K [R \ T]$, where K are called internal or intrinsic parameter and $[R \ T]$ are called the external or extrinsic parameters; the former are related to the intrinsic physical properties of the camera (e.g. focal length, etc...) where the latter are related to pose and location of the camera; Notice that these two set of parameters are nicely separated in this formulation.

- We added world information
 - Transformation from world reference system to the camera one
 - Extrinsic parameters of a camera only depend on its location and orientation
 - If I “move” the camera they need to be computed again
- $[RT] \rightarrow$ **Camera Extrinsic Matrix**

La componente matriciale RT è la matrice degli estrinseci. Non dipende dai parametri della telecamera ma solo dove si trova e come è orientata la telecamera stessa

- 11 degrees of freedom (5 + 3 + 3)



$$P'_{3 \times 1} = M_{3 \times 4} P_w = K_{3 \times 3} \begin{bmatrix} R & T \end{bmatrix}_{3 \times 4} P_{w4 \times 1}$$

Courtesy: Silvio Savarese

11 gradi di libertà ovvero per sapere dove un punto nel mondo va a finire nella mia immagine devo conoscere 11 parametri.

We have rewritten here Eq. 11 again to highlight the dimensionality of the matrices and vectors; For instance, $M_{3 \times 4}$ means that the matrix M has 3 rows and 4 columns. How many degrees of freedom do M have?

- 5 from K (as we have seen earlier).
- 3 from R (a rotation matrix in 3D has 3 degree of freedom).
- 3 from the translation vector T . In total 11! Eq. 11 describes the projective transformation and M is a projective matrix.



$$P'_{3 \times 1} = M P_w = K_{3 \times 3} \begin{bmatrix} R & T \end{bmatrix}_{3 \times 4} P_w_{4 \times 1} \quad M = \begin{bmatrix} \mathbf{m}_1 \\ \mathbf{m}_2 \\ \mathbf{m}_3 \end{bmatrix}$$

$$= \begin{bmatrix} \mathbf{m}_1 \\ \mathbf{m}_2 \\ \mathbf{m}_3 \end{bmatrix} P_w = \begin{bmatrix} \mathbf{m}_1 P_w \\ \mathbf{m}_2 P_w \\ \mathbf{m}_3 P_w \end{bmatrix} \xrightarrow{E} \left(\frac{\mathbf{m}_1 P_w}{\mathbf{m}_3 P_w}, \frac{\mathbf{m}_2 P_w}{\mathbf{m}_3 P_w} \right) \quad [\text{Eq. 12}]$$

Courtesy: Silvio Savarese

I vai $\mathbf{m}_x$ mi rappresentano una riga 1×4 della mia M . In pratica le 3 coordinate omogenee del piano immagine le posso vedere come questi $\mathbf{m}_x$ moltiplicati per il punto immagine (4×1). Per ritornare alle coordinate cartesiane divido le prime due coordinate omogenee per la terza. (eq. 12)

The mapping between P_w and P' is in homogenous coordinates. If we want to obtain P' in Euclidean coordinates (which corresponds to our actual measurements in the image), we need to change coordinates from homogenous to Euclidean by dropping the last coordinate of P' and dividing the first two coordinates by the last coordinate. This operation is illustrated in Eq. 12. Here **$\mathbf{m}_1$, $\mathbf{m}_2$ and $\mathbf{m}_3$ are the first, second and third row vectors of M , respectively. Thus, each of the $\mathbf{m}_i$ is 1×4 vector.** As Eq. 12 shows, the final outcome of the most generic formulation of a projective transformation in Euclidean coordinates is a rather complex (and non linear) intermix of the intrinsic and extrinsic parameters. However, by deriving this mapping in the homogenous coordinates, we managed to have a good understanding of the geometrical properties of the problem where intrinsic and extrinsic parameters were separated.

Perspective transformation properties



- Points become.... Points!
- Lines become... Lines!
- Far away objects are smaller (divide by z)





- Angles are not preserved
- Parallel lines intersect!
 - In the so-called **vanishing point**



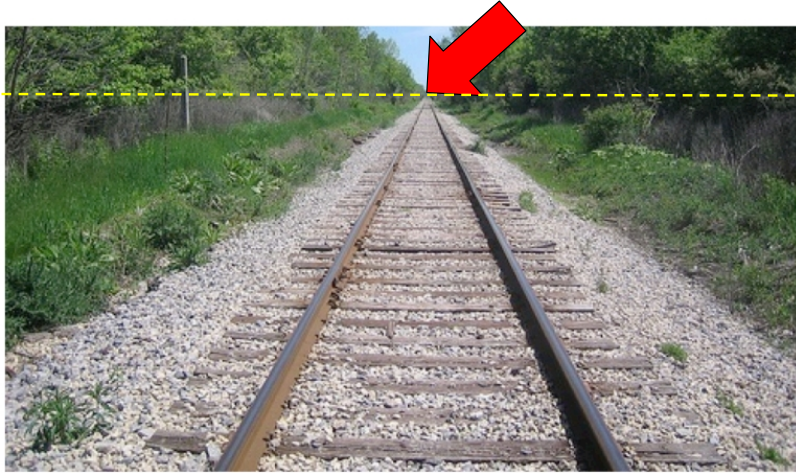
Linee parallele nel mondo 3D non è detto che lo rimangano dopo la proiezione. In generale si intersecheranno tutte su una ben precisa linea detta linea di fuga (o anche linea dell'orizzonte).

Two parallel lines in 3D are no longer necessarily parallel after a projective transformation; that is, in general parallel lines in 3D intersect at one point after a projective transformation. As we will discuss in more details in lecture 4, this point is called vanishing point. An example of vanishing point is illustrated above.

Perspective transformation properties



- Parallel lines that lie in the same plane have vanishing points on a line
- The Horizon (Vanishing Line)

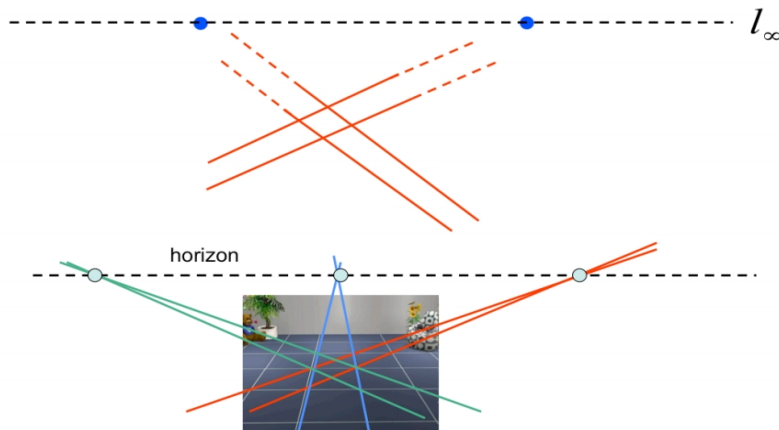


-By considering multiple pairs of parallel lines on the same plane (the ground plane in this example), we obtain a set of vanishing points which have the property of being collinear. The line that passes through these vanishing points is called horizon.

Perspective transformation properties



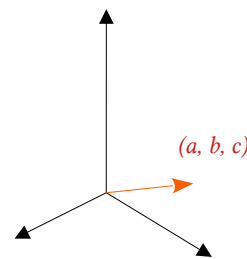
- Parallel lines that lie in the same plane have vanishing points on a line
- The Horizon (Vanishing Line)



-By considering multiple pairs of parallel lines on the same plane (the ground plane in this example), we obtain a set of vanishing points which have the property of being collinear. The line that passes through these vanishing points is called horizon.

- A 3D straight line can be expressed as:

$$\begin{cases} x(t) = x_0 + at \\ y(t) = y_0 + bt \\ z(t) = z_0 + ct \end{cases} \longrightarrow \begin{cases} x'(t) = f \frac{(x_0 + at)}{(z_0 + ct)} \\ y'(t) = f \frac{(y_0 + bt)}{(z_0 + ct)} \end{cases}$$



$$\begin{cases} \lim_{t \rightarrow \pm\infty} x'(t) = f \frac{a}{c} \\ \lim_{t \rightarrow \pm\infty} y'(t) = f \frac{b}{c} \end{cases}$$

- x_0, y_0, z_0 represent a point of the line, where t is the distance from that point.
- a, b, c then encode the direction
- Parallel lines have the same a, b , and c params!
- Points at infinity are not at infinity in the image

Questa è la dimostrazione

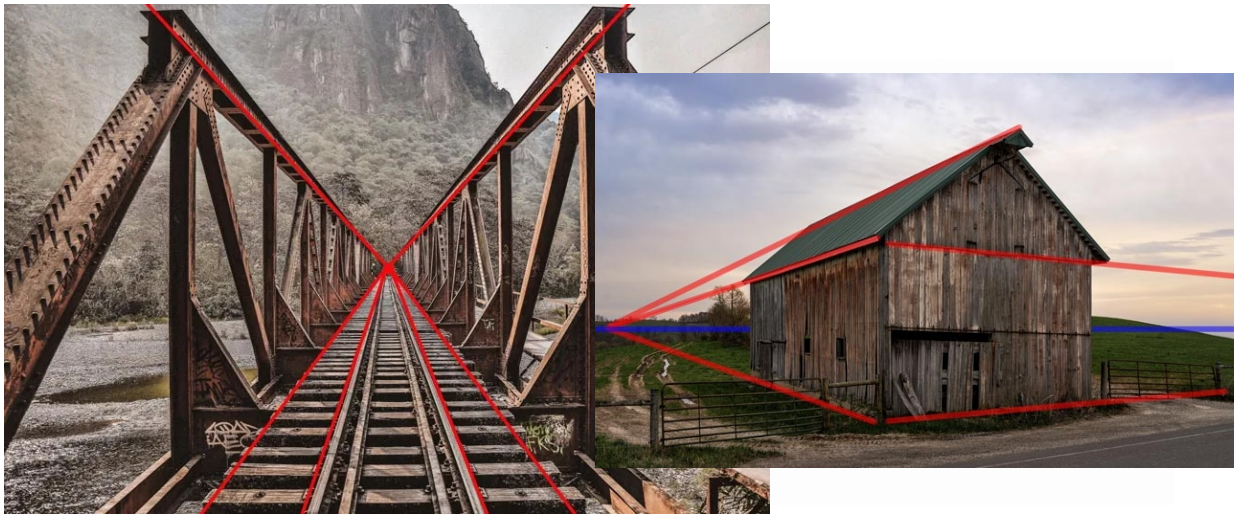
Fissato un punto x_0, y_0, z_0 da cui la retta passa t è il modulo del vettore che parte da quel punto ed identificato come direzione da a, b, c allineato alla retta

Quindi l'equazione di una retta nel mondo 3D la posso vedere come un sistema di 3 funzioni $f(t)$

Proiettiamo queste funzioni (usiamo la proiezione "semplice" vista nelle prime slide, ma il risultato sarebbe comunque lo stesso anche usando una M completa).

Otengo 3 funzioni dette omografiche che è noto abbiano un asintoto orizzontale (nel piano t).

Rette con stessi a, b, c sono parallele nel mondo (camberanno i punti x_0, y_0, z_0) ma per t tendente a $\pm\infty$ finiscono nello stesso punto del piano immagine!



Esempi di rette parallele nel mondo.

Occhio che la linea di fuga non è quella per tutte le rette parallele...

Esempio. Dove si intersecano le rette delle assi verticali del casotto o dei piloni diagonali del ponte? Chiaramente non dove si intersecano le altre...

Motivo per cui parlare di linea dell'orizzonte non è correttissimo, sarebbe meglio usare linea di fuga.



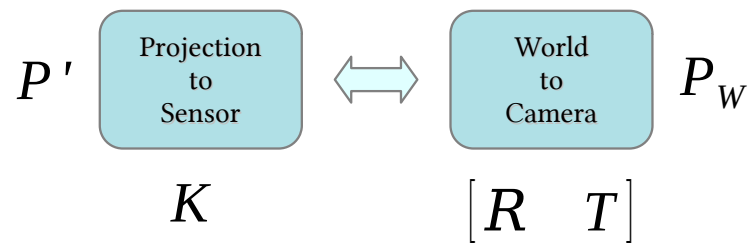
$$M = K \cdot \begin{bmatrix} I & 0 \end{bmatrix} \cdot E = \overset{\text{Intrinsic}}{\boxed{K}} \cdot \underset{\text{Extrinsic}}{\boxed{\begin{bmatrix} R & T \end{bmatrix}}} \in R^{3 \times 4}$$

$$\underset{\substack{\uparrow \\ \text{2D Homogeneous}}}{P'} = \underset{\substack{\uparrow \\ \text{Model the perspective} \\ \text{transformation from 3D to 2D}}}{M} \cdot \underset{\substack{\nwarrow \\ \text{3D Homogeneous}}}{P_w} = K \cdot \begin{bmatrix} R & T \end{bmatrix} \cdot P_w$$

RECAP

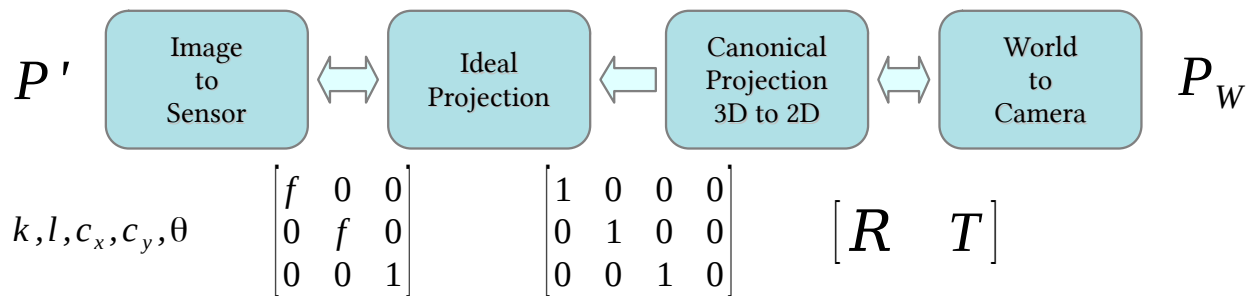


- We can see this transformation as the combination of
 - World to camera
 - Projection to sensor



- The process can be further decomposed
- Also image to sensor can be further decomposed
 - Euclidean, affine, and general

Arrows have a meaning!



Voi direte, ma a che serve? In realtà la cosiddetta telecamera canonica viene usata nel modello epipolare